

FC7xxx PWM User Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.3	2023/10/20	Updated for MCAL V0.3.0
0.4	2023/11/22	Updated for MCAL V0.4.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 PWM Introduction

1.1 Requirements

The design of this module follows the specifications of the PWM driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_PWMDriver*.

1.2 Hardware Summary

The PWM driver provides logical channels which are configured by hardware including Flexible Timer Unit (FTU).

The PWM driver provides several services including initializing channels, de-initializing channels, forcing the output to idle, enabling/disabling the trigger out and others.

The PWM driver implements a maximum of 96 channels on 1 FC7300 peripheral and implements a maximum of 64 channels on 1 FC7240 peripheral

96 channels are implemented on 12 FTU modules(FC7300).

64 channels are implemented on 8 FTU modules(FC7240).

FTU channels

A maximum of eight 16-bit counters on each module FTU (0~11).

For FC7240, FTU1 and FTU2 support 24-bit counter width, and other FTUs support 16-bit counter width.

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 1 **SWS_Pwm_50075**

Description	Pwm.c shall include Pwm.h, Pwm_MemMap.h, Det.h and SchM_Pwm.h.
Rejection Reason	They are included in underlying logic files.

Rejected Requirement 2 **SWS_Pwm_70075**

Description	Pwm_Irq.c shall include Pwm_MemMap.h and Pwm.h
Rejection Reason	Pwm_Irq.c has not been implemented.

Rejected Requirement 3 **SWS_Pwm_00162**

Description	The PWM Driver shall support synchronous and asynchronous power state transitions, depending on the value of the configuration parameter PwmPowerStateAsynchTransitionMode.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 4 **SWS_Pwm_00164**

Description	In case the configuration parameter PwmPowerStateAsynchTransitionMode is set to TRUE, the preparation process shall continue in background after the relative API returns and its completion shall be notified by means of the configured callback.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 5 **SWS_Pwm_00189**

Description	Service name	Pwm_Main_PowerTransitionManager
	Syntax	void Pwm_Main_PowerTransitionManager(void)
	Service ID[hex]	0x0d
	Description	This API is cyclically called and supervises the power state transitions, checking for the readiness of the module and issuing the callbacks IoHwAb_Pwm_NotifyReadyForPowerState<Mode> (see PwmPowerStateReadyCbRef configuration parameter).
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.	

Rejected Requirement 6 **SWS_Pwm_00190**

Description	This API executes any non-immediate action needed to finalize a power state transition requested by Pwm_PreparePowerState().
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 7 SWS_Pwm_00191

Description	The rate of scheduling shall be defined by PWM MainSchedulePeriod and shall be variable, as the function only needs to be called if a transition has been requested.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 8 SWS_Pwm_00192

Description	This API shall also issue callback notifications to the eventually registered users (IoHwAbs) as configured, only in case the asynch mode is chosen.?()
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 9 SWS_Pwm_00193

Description	In case the PWM module is not initialized, this function shall simply return without any further elaboration. This is needed to avoid to elaborate uninitialized variables. No DET error shall be entered, because this condition can easily be verified during the startup phase (tasks started before the initialization is complete).Rationale: during the startup phase it can happen that the OS already schedules tasks, which call main functions, while some modules are not initialised yet. This is no real error condition, although need handling, i.e. returning without execution. Although the transition state monitoring functionality is mandatory, the implementation of this API is optional, meaning that if the HW allows for other ways to deliver notification and watch the transition state the implementation of this function can be skipped.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 10 SWS_Pwm_00198

Description	Service name	IoHwAb_Pwm_NotifyReadyForPowerState<#Mode>
	Syntax	void IoHwAb_Pwm_NotifyReadyForPowerState<#Mode>(void)
	Service ID[hex]	0x60
	Sync/Async:	Synchronous
	Reentrancy	Non Reentrant
	Parameters (in)	None
	Parameters (inout)	None
	Parameters (out)	None
	Return value	None
	Description	The API shall be invoked by the PWM Driver when the requested power state preparation for mode <#Mode> is completed.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.	

Rejected Requirement 11 SWS_Pwm_00199

Description	In case the PWM Driver is configured to support power state management with asynchronous
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	transitions, this API shall be called to signal completion of the power transition preparation phase to the IoHwAbs module. This is a callback, this API is to be implemented in the IoHwAbs component.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.

Rejected Requirement 12 SWS_Pwm_00153

Description	These requirements are not applicable to this specification. (SRS_BSW_00159, SRS_BSW_00167, SRS_BSW_00170, SRS_BSW_00419, SRS_BSW_00383, SRS_BSW_00375, SRS_BSW_00416, SRS_BSW_00168, SRS_BSW_00423, SRS_BSW_00424, SRS_BSW_00425, SRS_BSW_00426, SRS_BSW_00427, SRS_BSW_00428, SRS_BSW_00429, BSW00431, SRS_BSW_00432, SRS_BSW_00433, BSW00434, SRS_BSW_00417, SRS_BSW_00161, SRS_BSW_00162, SRS_BSW_00005, SRS_BSW_00415, SRS_BSW_00164, SRS_BSW_00325, SRS_BSW_00326, SRS_BSW_00342, SRS_BSW_00160, SRS_BSW_00007, SRS_BSW_00300, SRS_BSW_00413, SRS_BSW_00347, SRS_BSW_00305, SRS_BSW_00307, SRS_BSW_00310, SRS_BSW_00373, SRS_BSW_00327, SRS_BSW_00335, SRS_BSW_00350, SRS_BSW_00408, SRS_BSW_00410, SRS_BSW_00348, SRS_BSW_00353, SRS_BSW_00361, SRS_BSW_00301, SRS_BSW_00302, SRS_BSW_00328, SRS_BSW_00312, SRS_BSW_00006, SRS_BSW_00357, SRS_BSW_00377, SRS_BSW_00304, SRS_BSW_00355, SRS_BSW_00378, SRS_BSW_00306, SRS_BSW_00308, SRS_BSW_00309, SRS_BSW_00371, SRS_BSW_00358, SRS_BSW_00414, SRS_BSW_00376, SRS_BSW_00359, SRS_BSW_00360, SRS_BSW_00329, SRS_BSW_00330, SRS_BSW_00331, SRS_BSW_00009, SRS_BSW_00401, SRS_BSW_00172, SRS_BSW_00010, SRS_BSW_00333, SRS_BSW_00003, SRS_BSW_00341, SRS_BSW_00334, SRS_SPAL_12267, SRS_SPAL_12461, SRS_SPAL_12462, SRS_SPAL_12463, SRS_SPAL_12068, SRS_SPAL_12069, SRS_SPAL_12169, SRS_SPAL_12075, SRS_SPAL_12064, SRS_SPAL_12067, SRS_SPAL_12077, SRS_SPAL_12078, SRS_SPAL_12092, SRS_SPAL_12265, SRS_Pwm_12379)
Rejection Reason	It is not a requirement.

Rejected Requirement 13 ECUC_Pwm_00143

Description	SWS Item	ECUC_Pwm_00143		
	Name	PwmPowerStateAsynchTransitionMode		
	Parent Container	PwmGeneral		
	Description	Enables / disables support of the PWM Driver to the asynchronous power state transition.		
	Multiplicity	0..1		
	Type	EcucBooleanParamDef		
	Default value	false		
	Post-Build Variant Multiplicity	false		
	Post-Build Variant Value	false		
	Multiplicity Configuration Class	Pre-compile time	X	All Variants
Link time		--		
Post-build time		--		

	Value Configuration Class	Pre-compile time	X	All Variants
		Link time	--	
		Post-build time	--	
	Scope / Dependency	scope: local dependency: This parameter shall only be configured if the parameter PwmLowPowerStatesSupport is set to true.		
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.			

Rejected Requirement 14 ECUC_Pwm_00144

Description	SWS Item	ECUC_Pwm_00144
	Container Name	PwmPowerStateConfig
	Description	Each instance of this parameter defines a power state and the callback to be called when this power state is reached.
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.	

Rejected Requirement 15 ECUC_Pwm_00146

Description	SWS Item	ECUC_Pwm_00146		
	Name	PwmPowerState		
	Parent Container	PwmPowerStateConfig		
	Description	Each instance of this parameter describes a different power state supported by the PWM HW. It should be defined by the HW supplier and used by the PWM Driver to reference specific HW configurations which set the PWM HW module in the referenced power state.		
	Multiplicity	1		
	Type	EcucIntegerParamDef (Symbolic Name generated for this parameter)		
	Range	0 .. 18446744073709551615		
	Post-Build Variant Multiplicity	false		
	Post-Build Variant Value	false		
	Value Configuration Class	Pre-compile time	X	All Variants
		Link time	--	
		Post-build time	--	
	Scope / Dependency	scope: local dependency: This parameter shall only be configured if the parameter PwmLowPowerStatesSupport is set to true.		
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.			

Rejected Requirement 16 ECUC_Pwm_00145

Description	SWS Item	ECUC_Pwm_00145
	Name	PwmPowerStateReadyCbRef
	Parent Container	

	Description	Each instance of this parameter contains a reference to a power mode callback defined in a CDD or IoHwAbs component.		
	Multiplicity	1		
	Type	EcucFunctionNameDef		
	Default value	--		
	maxLength	--		
	minLength	--		
	regularExpression	--		
	Post-Build Variant Value	false		
	Value Configuration Class	Pre-compile time	X	All Variants
		Link time	--	
		Post-build time	--	
	Scope / Dependency	scope: local dependency: This parameter shall only be configured if the parameter PwmLowPowerStatesSupport is set to true.		
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.			
Rejection Reason	Synchronous and asynchronous power state transitions are not supported.			

Rejected Requirement 17 SWS_Pwm_00093

Description	The users of the PWM module shall not call the function Pwm_Init during a running operation.
Rejection Reason	Depending on users.

Rejected Requirement 18 SWS_Pwm_00116

Description	The PWM module's environment shall not call any function of the PWM module before having called Pwm_Init.
Rejection Reason	Depending on users.

Rejected Requirement 19 SWS_Pwm_00105

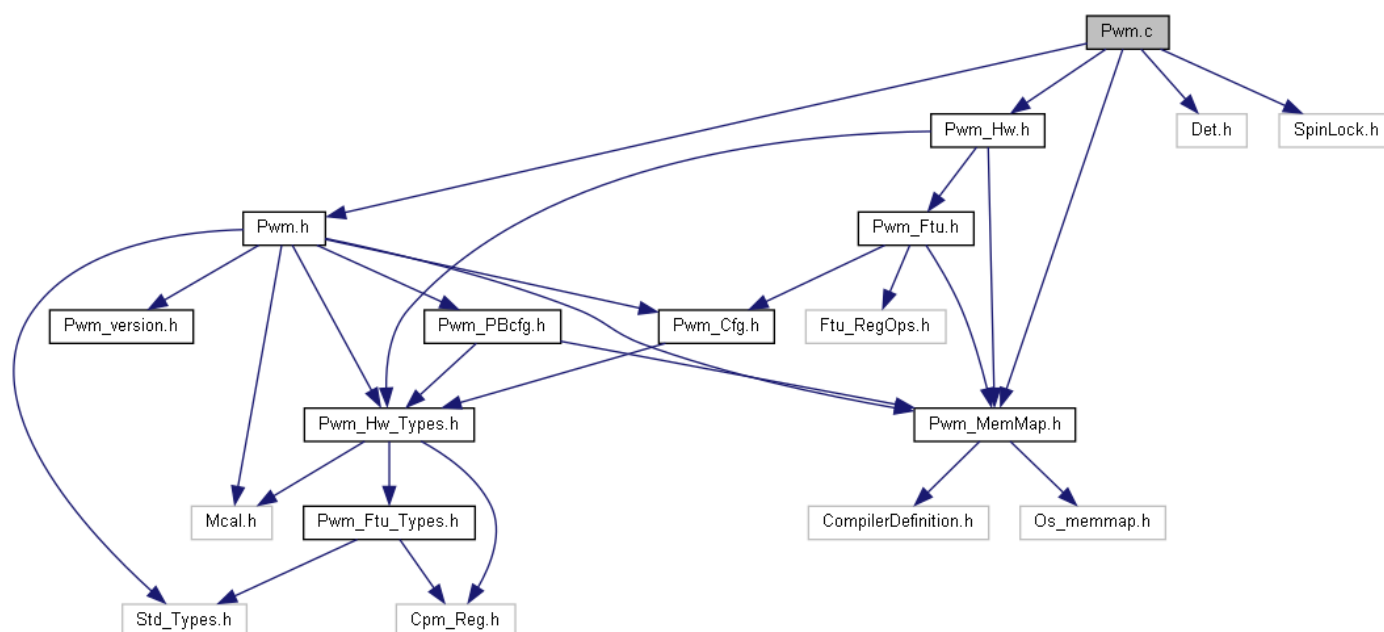
Description	Service name:	Pwm_Notification_<#Channel>
	Syntax:	void Pwm_Notification_<#Channel>(void)
	Sync/Async:	Synchronous
	Reentrancy:	PWM user implementation dependant
	Parameters (in):	None
	Parameters (inout):	None
	Parameters (out):	None
	Return value:	None
	Description:	The Pwm module shall call the function Pwm_Notification_<#Channel> accordingly to the last call of Pwm_EnableNotification for channel <#Channel>.
Rejection Reason	The user notification implemptation defined by user self.	

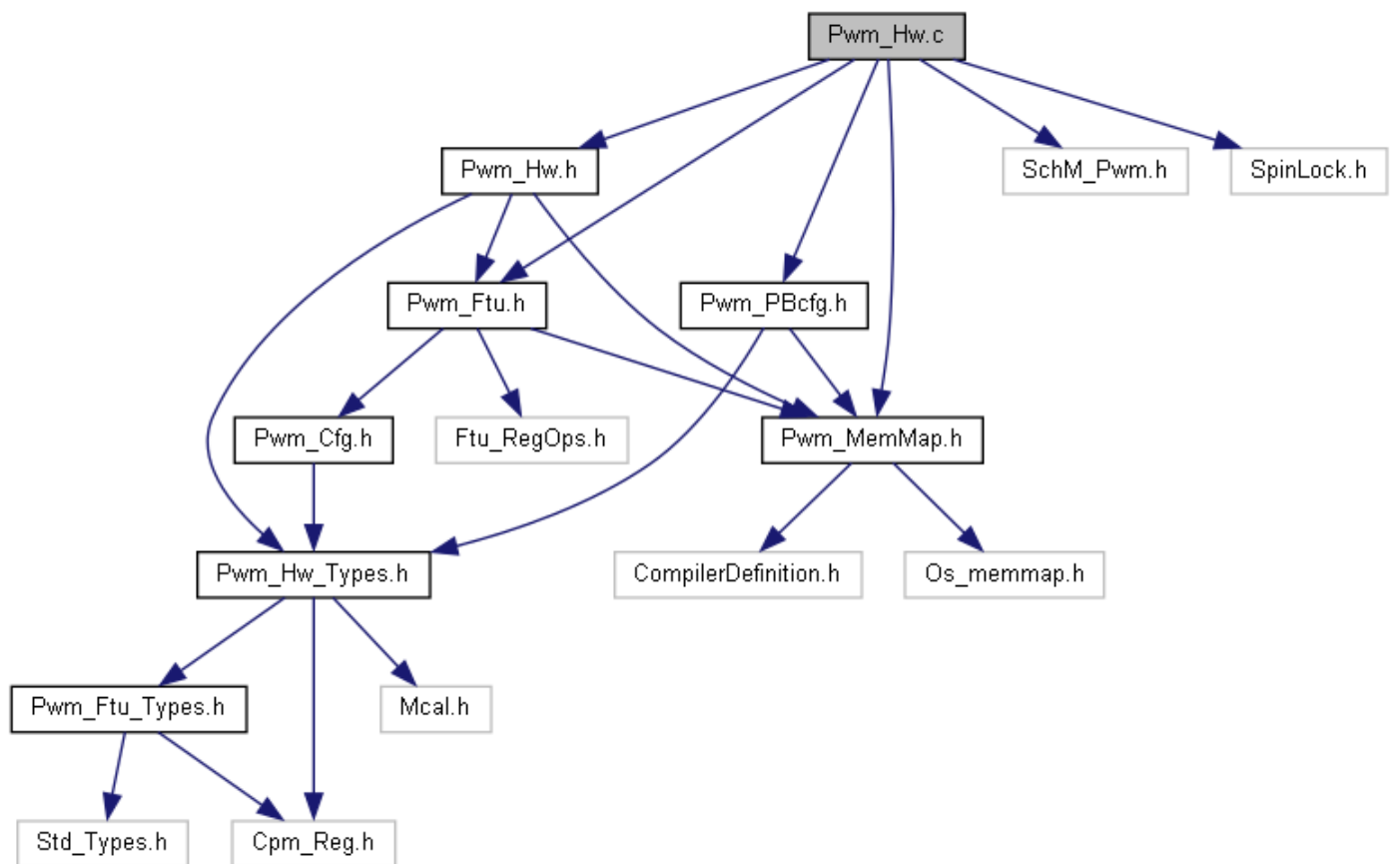
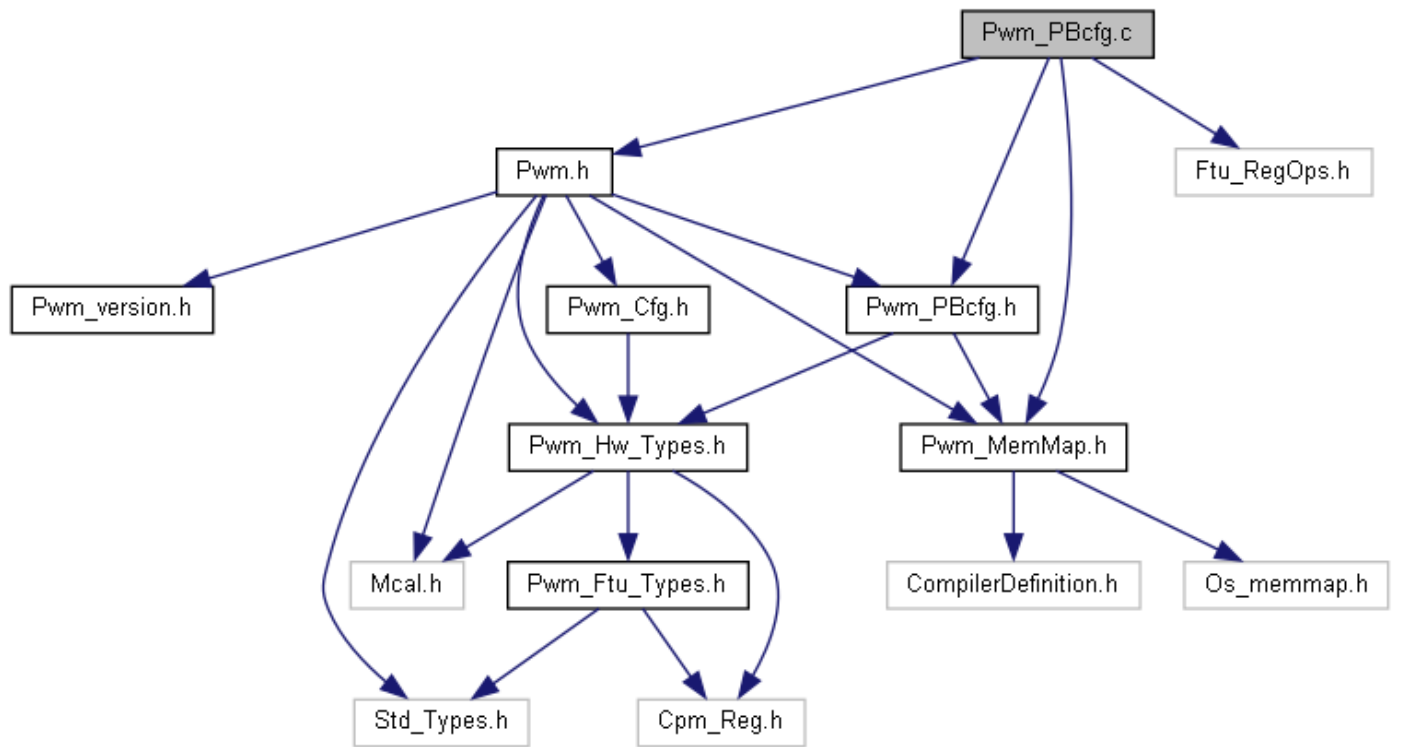
Rejected Requirement 20 SWS_Pwm_00025	
Description	The Pwm module shall call the function Pwm_Notification_<#Channel> accordingly to the last call of Pwm_EnableNotification and Pwm_DisableNotification for channel <#Channel>.
Rejection Reason	The user notification implementation defined by user self.

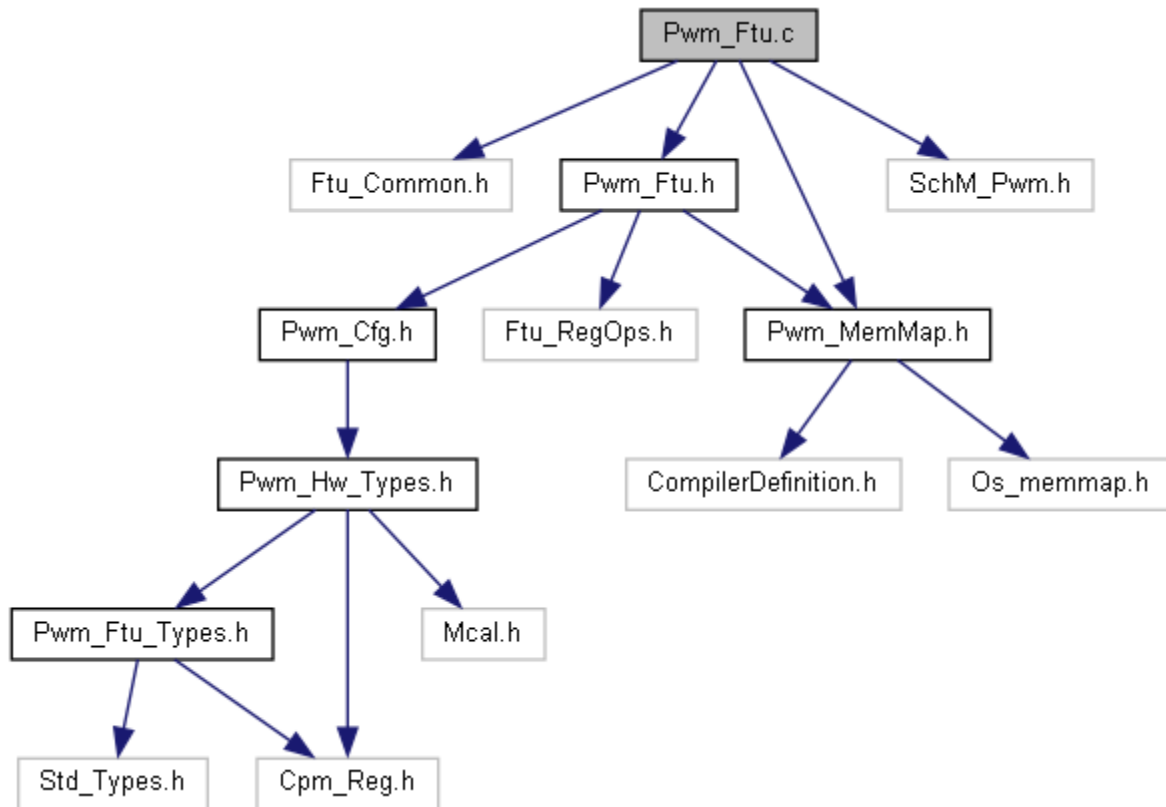
Rejected Requirement 21 SWS_Pwm_00185	
Description	In case the target power state is the same as the current one, no action is executed and the API returns immediately with an E_OK result. The responsibility of the preconditions is left to the environment. In case development error reporting is activated.
Rejection Reason	This requirement conflicts with requirement SWS_Pwm_00183

Rejected Requirement 22 SWS_Pwm_00188	
Description	The API shall report the DET error PWM_E_TRANSITION_NOT POSSIBLE in case the requested power state cannot be directly reached from the current power state. All asynchronous operation needed to reach the target power state can be executed in background in the context of Pwm_Main_PowerTransitionManager.
Rejection Reason	This requirement is not allowed in current hardware

2.2 File Structure







2.3 Macros

2.3.1 Macros in Pwm.h

- **#define PWM_DUTY_CYCLE_100** ((uint16)0x8000U)
- **#define PWM_INIT_CORE_USED_VALID** ((uint8)0x01U)
- **#define PWM_INIT_CORE_USED_INVALID** ((uint8)0x00U)
- **#define PWM_SPIN_LOCK_TIMEOUT** (0xFFFFU)
The timeout for acquiring a spin lock.
- **#define PWM_E_INIT_FAILED** (0x10U)
Initialization function called failed.
- **#define PWM_E_UNINIT** (0x11U)
Function called without module initialization.
- **#define PWM_E_PARAM_CHANNEL** (0x12U)|
Function called with invalid channel parameter.
- **#define PWM_E_PERIOD_UNCHANGEABLE** (0x13U)
Set period function called with the channel which is unchangeable.

- **#define PWM_E_ALREADY_INITIALIZED (0x14U)**
Initialization function called again.
- **#define PWM_E_PARAM_POINTER (0x15U)**
Function called with invalid pointer parameter.
- **#define PWM_E_NOT_DISENGAGED (0x16U)**
Set power state function called when the channel is disengaged.
- **#define PWM_E_POWER_STATE_NOT_SUPPORTED (0x17U)**
Power state function is not supported.
- **#define PWM_E_TRANSITION_NOT_POSSIBLE (0x18U)**
The requested power state cannot be directly reached from the current power state.
- **#define PWM_E_PERIPHERAL_NOT_PREPARED (0x19U)**
The HW unit has not been previously prepared for the target power state by use of the API `Pwm_PreparePowerState()`.
- **#define PWM_E_PARAM_NOTIFICATION (0x30U)**
The pointer to notification function is error.
- **#define PWM_E_PARAM_NOTIFICATION_NULL (0x31U)**
The pointer to notification function is null.
- **#define PWM_E_DUTYCYCLE_RANGE (0x32U)**
Invalid duty cycle range value.
- **#define PWM_E_PARAM_CONFIG (0x40U)**
Generated when the requested resource is configured to be unavailable on the current core.
- **#define PWM_INIT_ID (0x00U)**
Initialization function ID.
- **#define PWM_DEINIT_ID (0x01U)**
De-initialization function ID.
- **#define PWM_SETDUTYCYCLE_ID (0x02U)**
Set duty cycle function ID.
- **#define PWM_SETPERIODANDDUTY_ID (0x03U)**
Set period and duty cycle function ID.

- **#define PWM_SETOUTPUTTOIDLE_ID** (0x04U)
Set output to idle function ID.
- **#define PWM_GETOUTPUTSTATE_ID** (0x05U)
Get output state function ID.
- **#define PWM_DISABLENOTIFICATION_ID** (0x06U)
Disable notification function ID.
- **#define PWM_ENABLENOTIFICATION_ID** (0x07U)
Enable notification function ID.
- **#define PWM_GETVERSIONINFO_ID** (0x08U)
Get version function ID.
- **#define PWM_SETPOWERSTATE_ID** (0x09U)
Set power state function ID.
- **#define PWM_GETCURRENTPOWERSTATE_ID** (0x0AU)
Get current power state function ID.
- **#define PWM_GETTARGETPOWERSTATE_ID** (0x0BU)
Get target power state function ID.
- **#define PWM_PREPAREPOWERSTATE_ID** (0x0CU)
Prepare power state function ID.
- **#define PWM_GETCHANNELSTATE_ID** (0x20U)
Get channel state function ID.
- **#define PWM_ENABLETRIGGER_ID** (0x21U)
Enable trigger function ID.
- **#define PWM_DISABLETRIGGER_ID** (0x22U)
Disable trigger function ID.
- **#define PWM_MASKOUTPUT_ID** (0x23U)
Mask output function ID.
- **#define PWM_UNMASKOUTPUT_ID** (0x24U)
Unmask output function ID.

- #define **PWM_STARTGLOBALTIME_ID** (0x30U)
Enable FTU global time base function ID.
- #define **PWM_STOPGLOBALTIME_ID** (0x31U)
Disable FTU global time base function ID.

2.3.2 Macros in Pwm_version.h

- #define **PWM_AR_RELEASE_MAJOR_VERSION** 4
- #define **PWM_AR_RELEASE_MINOR_VERSION** 6
- #define **PWM_AR_RELEASE_REVISION_VERSION** 0
- #define **PWM_SW_MAJOR_VERSION** 0
- #define **PWM_SW_MINOR_VERSION** 6
- #define **PWM_SW_PATCH_VERSION** 0
- #define **PWM_VENDOR_ID** 174
- #define **PWM_MODULE_ID** 121

2.3.3 Macros in Pwm_Cfg.h

- #define **PWM_CFG_AR_RELEASE_MAJOR_VERSION** 4
- #define **PWM_CFG_AR_RELEASE_MINOR_VERSION** 6
- #define **PWM_CFG_AR_RELEASE_REVISION_VERSION** 0
- #define **PWM_CFG_MODULE_ID** 121
- #define **PWM_CFG_SW_MAJOR_VERSION** 0
- #define **PWM_CFG_SW_MINOR_VERSION** 6
- #define **PWM_CFG_SW_PATCH_VERSION** 0
- #define **PWM_CFG_VENDOR_ID** 174
- #define **PWM_DE_INIT_API** (STD_ON)
Switch to indicate that Pwm_DeInit API is supported.

- **#define PWM_DEV_ERROR_DETECT (STD_ON)**
Switch for enabling the development error detection.
- **#define PWM_DUTY_PERIOD_UPDATED_ENDPERIOD (STD_ON)**
Switch for enabling the update of the period parameter at the end of the current period.
- **#define PWM_E_PERIODVALUE (0x1AU)**
Symbolic Name for period det error.
- **#define PWM_ENABLE_MASKING_OPERATIONS (STD_OFF)**
Switch for enabling MaskOutput API.
- **#define PWM_ENABLE_USER_MODE_SUPPORT (STD_OFF)**
Support for User mode. If this parameter has been configured to 'TRUE' the PWM driver code can be executed from both supervisor and user mode.
- **#define PWM_MULTICORE_ENABLED (STD_ON)**
Enable/Disable multicore feature for PWM driver.
- **#define PWM_FAULT_SUPPORTED (STD_ON)**
Switch for enabling the fault functionality.
- **#define PWM_FTU_0_ISR_USED (STD_ON)**
ISRs configured for PWM channels FTU module.
- **#define PWM_FTU_CHANNELS_NO (PWM_FTU_MODULE_CHANNELS_NO * PWM_FTU_MODULE_NO)**
- **#define PWM_FTU_MODULE_CHANNELS_NO (8U)**
- **#define PWM_FTU_MODULE_FAULT_NO (4U)**
- **#define PWM_FTU_MODULE_NO (6U)**
- **#define PWM_FTU_USED (STD_ON)**
Define to indicate FTUis used.
- **#define PWM_GET_CHANNEL_STATE_API (STD_ON)**
Switch to indicate that Pwm_GetChannelState API is supported.
- **#define PWM_GET_OUTPUT_STATE_API (STD_ON)**
Switch to indicate that Pwm_GetOutputState API is supported.
- **#define PWM_HW_CHANNELS_NO_U8 ((uint8)0U + PWM_FTU_CHANNELS_NO)**

- **#define PWM_HW_MODULES_CFG_U8** (PWM_FTU_MODULE_NO)
- **#define PWM_INDEX** (0U)
Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0. Not used in the current implementation.
- **#define PWM_MAX_PERIOD** (0xFFFFU)
Defines the maximum value of the period - this value is hardware/platform dependent.
- **#define PWM_NOTIFICATION_SUPPORTED** (STD_ON)
Switch to indicate that the notifications are supported.
- **#define PWM_POWER_STATE_ASYNC_MODE_SUPPORTED** (STD_OFF)
- **#define PWM_POWER_STATE_SUPPORTED** (STD_OFF)
Switch to enable that power state mode is supported.
- **#define PWM_PRECOMPILE_SUPPORT** (STD_ON)
PWM Pre Compile Switch.
Pre-compile configuration constants.
- **#define PWM_PROCESSNOTIFICATION_ID** (0x0AU)
API service ID.
- **#define PWM_SET_DUTY_CYCLE_API** (STD_ON)
Switch to indicate that Pwm_SetDutyCycle API is supported.
- **#define PWM_SET_OUTPUT_TO_IDLE_API** (STD_ON)
Switch to indicate that Pwm_SetOutputToIdle API is supported.
- **#define PWM_SET_PERIOD_AND_DUTY_API** (STD_ON)
Switch to indicate that Pwm_SetPeriodAndDuty API is supported.
- **#define PWM_TRIGGER_OUT_API** (STD_ON)
Switch for enabling Pwm_EnableTriggerOut()/Pwm_DisableTriggerOut() API.
- **#define PWM_VERSION_INFO_API** (STD_ON)
Switch to indicate that Pwm_GetVersionInfo API is supported.
- **#define PwmChannel_0** ((Pwm_ChannelType)0U)
Symbolic Names for configured channels - legacy.
- **#define PwmConf_PwmChannelConfigSet_PwmChannel_0** ((Pwm_ChannelType)0U)
Symbolic Names for configured channels - ecuc 2108 compliant.

2.4 Enums

2.4.1 Enumerations in Pwm_Hw_Types.h

2.4.1.1 Pwm_OutputStateType

Enumeration	Pwm_OutputStateType	
Description	Output state of a PWM channel.	
Values	Value	Description
	PWM_LOW = 0	PWM level is logic low.
	PWM_HIGH = 1	PWM level is logic high.

2.4.1.2 Pwm_EdgeNotificationType

Enumeration	Pwm_EdgeNotificationType	
Description	Definition of the type of edge notification of a PWM channel.	
Values	Value	Description
	PWM_RISING_EDGE = 1	A notification will be generated on the rising edge.
	PWM_FALLING_EDGE = 2	A notification will be generated on the falling edge.
	PWM_BOTH_EDGES = 3	A notification will be generated on any state transition.

2.4.1.3 Pwm_PowerStateType

Enumeration	Pwm_PowerStateType	
Description	Power state currently active or set as target power state.	
Values	Value	Description
	PWM_FULL_POWER = 0	PWM full power mode.
	PWM_LOW_POWER = 1	PWM low power mode.
	PWM_NODEFINE_POWER = 2	PWM no define power mode.

2.4.1.4 Pwm_PowerStateRequestResultType

Enumeration	Pwm_PowerStateRequestResultType	
Description	Result of the requests related to power state transitions.	
Values	Value	Description
	PWM_SERVICE_ACCEPTED = 0	Power state change executed.
	PWM_NOT_INIT = 1	Module not initialized.
	PWM_SEQUENCE_ERROR = 2	Wrong API call sequence.
	PWM_HW_FAILURE = 3	The HW module has a failure that prevents it to enter the required power state.
	PWM_POWER_STATE_NOT_SUPP = 4	The module does not support the requested power state.
	PWM_TRANS_NOT_POSSIBLE = 5	The module cannot transition directly from the current power state to the requested power state.

2.4.1.5 Pwm_ChannelClassType

Enumeration	Pwm_ChannelClassType	
Description	Defines the class of a PWM channel.	
Values	Value	Description
	PWM_VARIABLE_PERIOD = 0	The PWM channel has a variable period. The duty cycle and the period can be changed.
	PWM_FIXED_PERIOD = 1	The PWM channel has a fixed period. Only the duty cycle can be changed.
	PWM_FIXED_PERIOD_SHIFTED = 2	The PWM channel has a fixed shifted period. Impossible to change it (only if supported by hardware).

2.5 Typedefs

2.5.1 Typedefs in Pwm_Hw_Types.h

- `typedef uint8 Pwm_ChannelType.`
Numeric identifier of a PWM channel.
- `typedef uint8 Pwm_ModuleType`
PWM hardware instance type.
- `typedef Pwm_Ftu_ModuleConfigType Pwm_ModuleConfigType`
FTU configuration type to HW configuration type.
- `typedef Pwm_Ftu_InstanceType Pwm_InstanceType`
FTU instance type to HW instance type.
- `typedef Pwm_Ftu_ChannelConfigType Pwm_Hw_ChannelConfigType`
FTU channel configuration to HW channel configuration type.

2.5.2 Typedefs in Pwm_Ftu_Types.h

- `typedef uint8 Pwm_Ftu_ChannelType`
Ftu HW channel type.
- `typedef uint8 Pwm_Ftu_EdgeStateType`
Ftu TOF IRQ state type ID.
- `typedef uint32 Pwm_GtbGroupType`
PWM global time base group type
- `typedef void(*Pwm_Ftu_FaultNotifyType) (uint8 u8ModuleId, uint8 u8FaultChannel)`
Fault channel notification typedef.

- typedef void(* Pwm_Ftu_NotifyType) (void)
Channel FTU notification typedef.

2.5.3 Typedefs in Pwm_Cfg.h

- typedef uint16 Pwm_PeriodType
PWM Period type (the value of the period is platform dependent and thus configurable).

2.6 Structures

2.6.1 Pwm_ConfigType

Structure	Pwm_ConfigType
Description	This is the type of data structure containing the initialization data for the PWM driver.
Diagram	<pre> graph BT Pwm_ConfigType -- pPwmConfigChannels --> Pwm_ChannelConfigType Pwm_ConfigType -- pPwmConfigModules --> Pwm_Ftu_ModuleConfigType Pwm_ChannelConfigType -- tHwConfig --> Pwm_Ftu_ChannelConfigType </pre>
Data Fields	- const Pwm_ChannelConfigType * pPwmConfigChannels - const uint32 u32PwmChannelsCount Channel configurations pointer. - const Pwm_ModuleConfigType * pPwmConfigModules Channel configurations count. - const uint32 u32PwmModulesCount Module configurations pointer. More... - const Pwm_ChannelType * pHwToLogicChannelMap Module configurations count. - const Pwm_ModuleType * pHwToLogicInstanceMap Logic channel number to HW channel map.

2.6.2 Pwm_ChannelConfigType

Structure	Pwm_ChannelConfigType
Description	PWM channel configuration structure type

Diagram	<pre> classDiagram Pwm_ChannelConfigType < .. Pwm_Ftu_ChannelConfigType Pwm_ChannelConfigType ..> Pwm_Ftu_ChannelConfigType : tHwConfig </pre>
Data Fields	• const Pwm_InstanceType ePwmModuleInstance • const Pwm_ChannelType u8HwChannelId • const Pwm_ChannelClassType ePwmChannelClass • const Pwm_OutputStateType ePwmPolarity • const Pwm_OutputStateType ePwmIdleState • const Pwm_Hw_ChannelConfigType tHwConfig • const uint8 u8CoreId

2.6.3 Pwm_Ftu_ModuleConfigType

Structure	Pwm_Ftu_ModuleConfigType
Description	FTU common configuration structure type.
Diagram	N/A
Data Fields	• const Pwm_Ftu_InstanceType eInstance • const uint32 u32RegConf Configuration register value. • const uint32 u32RegSc Status and control register value. • const uint8 u8ExternalClk • const uint32 u32RegSync Synchronize register value. • const uint32 u32RegMode Mode selection register value. • const uint32 u32RegDeadtime Deadtime register value. • const Pwm_Ftu_AlignmentType eAlignment Channel alignment type • const uint32 u32RegChctrlVal Channel control register value. • const uint16 u16RegModVal Default period (mod) register value. • const uint32 u32RegFctrl Fault control register value. • const uint32 u32RegFltpol Fault polarity register value. • const uint32 u32RegPwmload PWM load register value. • const Pwm_Ftu_FaultNotifyType pfFaultNotification [2]

	Pointer to fault notification function. - const uint32 u32RegExtTrigg Alternate status and control register value. - const uint8 bModuleCoreUsed [CORE_NUM] Only useful if multicore is enabled. The core allocation involved in the FTU instance
--	--

2.7 API Functions

2.7.1 Functions in Pwm.h

2.7.1.1 void Pwm_DeInit (void)

Function	void Pwm_DeInit (void)
Description	PWM module De-initialization.
Diagram	N/A
Parameters	N/A
Returns	N/A

2.7.1.2 void Pwm_DisableNotification (Pwm_ChannelType ChannelNumber)

Function	void Pwm_DisableNotification (Pwm_ChannelType ChannelNumber)	
Description	Disable PWM channel notification.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
Returns	N/A	

2.7.1.3 void Pwm_DisableTriggerOut (Pwm_ChannelType ChannelNumber, boolean bReloadTrig)

Function	void Pwm_DisableTriggerOut (Pwm_ChannelType ChannelNumber, boolean bReloadTrig)	
Description	PWM disable trigger output.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
	bReloadTrig	Disable PWM channel reload trigger out.
Returns	N/A	

2.7.1.4 void Pwm_EnableNotification (Pwm_ChannelType ChannelNumber, Pwm_EdgeNotificationType Notification)

Function	void Pwm_EnableNotification (Pwm_ChannelType ChannelNumber, Pwm_EdgeNotificationType Notification)	
Description	Enable PWM channel notification.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
	Notification	PWM channel notification function pointer.

Returns	N/A
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2.7.1.5 void Pwm_EnableTriggerOut (Pwm_ChannelType ChannelNumber, boolean bReloadTrig)

Function	void Pwm_EnableTriggerOut (Pwm_ChannelType ChannelNumber, boolean bReloadTrig)	
Description	PWM enable trigger output.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
	bReloadTrig	Enable PWM channel reload trigger out.
Returns	N/A	

2.7.1.6 uint16 Pwm_GetChannelState (Pwm_ChannelType ChannelNumber)

Function	uint16 Pwm_GetChannelState (Pwm_ChannelType ChannelNumber)	
Description	Get PWM channel state.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
Returns	PWM channel state	

2.7.1.7 const Pwm_ConfigType* Pwm_GetConfigPtr (void)

Function	const Pwm_ConfigType* Pwm_GetConfigPtr (void)	
Description	PWM get configuration pointer.	
Diagram	N/A	
Parameters	N/A	
Returns	Configuration pointer	

2.7.1.8 Std_ReturnType Pwm_GetCurrentPowerState (Pwm_PowerStateType * CurrentPowerState, Pwm_PowerStateRequestResultType * Result)

Function	Std_ReturnType Pwm_GetCurrentPowerState (Pwm_PowerStateType * CurrentPowerState, Pwm_PowerStateRequestResultType * Result)	
Description	Get PWM current power state.	
Diagram	N/A	
Parameters	Parameter	Description
	CurrentPowerState	Get current power state.
	Result	Get current power state result.
Returns	Call function result	

2.7.1.9 Pwm_OutputStateType Pwm_GetOutputState (Pwm_ChannelType ChannelNumber)

Function	Pwm_OutputStateType Pwm_GetOutputState (Pwm_ChannelType ChannelNumber)	
Description	Get PWM channel output idle state.	
Diagram	N/A	

Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
Returns	PWM output state value.	

2.7.1.10 Std_ReturnType Pwm_GetTargetPowerState (Pwm_PowerStateType * TargetPowerState, Pwm_PowerStateRequestResultType * Result)

Function	Std_ReturnType Pwm_GetTargetPowerState (Pwm_PowerStateType * TargetPowerState, Pwm_PowerStateRequestResultType * Result)	
Description	Get PWM target power state.	
Diagram	N/A	
Parameters	Parameter	Description
	TargetPowerState	Get target power state.
	Result	Get target power state result.
Returns	Call function result	

2.7.1.11 void Pwm_GetVersionInfo (Std_VersionInfoType * pVersioninfo)

Function	void Pwm_GetVersionInfo (Std_VersionInfoType * pVersioninfo)	
Description	PWM get version information.	
Diagram	N/A	
Parameters	Parameter	Description
	pVersioninfo	PWM module version information value pointer.
Returns	N/A	

2.7.1.12 void Pwm_StartGlobalTime (const uint32 u32Group)

Function	void Pwm_StartGlobalTime (const uint32 u32Group)	
Description	Enable the FTU global time base function	
Diagram	N/A	
Parameters	Parameter	Description
	u32Group	FTU global time base Group number
Returns	N/A	

2.7.1.13 void Pwm_StopGlobalTime (const uint32 u32Group)

Function	void Pwm_StopGlobalTime (const uint32 u32Group)	
Description	Disable the FTU global time base function	
Diagram	N/A	
Parameters	Parameter	Description
	u32Group	FTU global time base Group number
Returns	N/A	

2.7.1.14 void Pwm_Init (const Pwm_ConfigType * ConfigPtr)

Function	void Pwm_Init (const Pwm_ConfigType * ConfigPtr)	
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Description	PWM module initialization.	
Diagram	N/A	
Parameters	Parameter	Description
	ConfigPtr	PWM configuration which configured from configuration tool or others.
Returns	N/A	

2.7.1.15 void Pwm_Main_PowerTransitionManager (void)

Function	void Pwm_Main_PowerTransitionManager (void)	
Description	PWM main power transition manger.	
Diagram	N/A	
Parameters	N/A	
Returns	N/A	

2.7.1.16 void Pwm_MaskOutputs (Pwm_ChannelType ChannelNumber)

Function	void Pwm_MaskOutputs (Pwm_ChannelType ChannelNumber)	
Description	PWM set mask output. It will force the channel to its inactive state.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	input PWM channel number
Returns	N/A	

2.7.1.17 void Pwm_Notification (Pwm_ChannelType u8ChannelIndex)

Function	void Pwm_Notification (Pwm_ChannelType u8ChannelIndex)	
Description	PWM notification.	
Diagram	N/A	
Parameters	Parameter	Description
	u8ChannelIndex	PWM channel map index
Returns	N/A	

2.7.1.18 Std_ReturnType Pwm_PreparePowerState (Pwm_PowerStateType PowerState, Pwm_PowerStateRequestResultType * Result)

Function	Std_ReturnType Pwm_PreparePowerState (Pwm_PowerStateType PowerState, Pwm_PowerStateRequestResultType * Result)	
Description	Prepare PWM power state.	
Diagram	N/A	
Parameters	Parameter	Description
	PowerState	Prepare PWM power state
	Result	Prepare PWM power state result
Returns	Call function result	

2.7.1.19 void Pwm_SetDutyCycle (Pwm_ChannelType ChannelNumber, uint16 DutyCycle)

Function	void Pwm_SetDutyCycle (Pwm_ChannelType ChannelNumber, uint16 DutyCycle)	
Description	Set PWM duty cycle.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	input PWM channel number
	DutyCycle	input PWM duty cycle value, the range is from 0x0000 to 0x8000
Returns	N/A	

2.7.1.20 void Pwm_SetOutputTogle (Pwm_ChannelType ChannelNumber)

Function	void Pwm_SetOutputTogle (Pwm_ChannelType ChannelNumber)	
Description	Set PWM channel output idle state.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
Returns	N/A	

2.7.1.21 void Pwm_SetPeriodAndDuty (Pwm_ChannelType ChannelNumber, Pwm_PeriodType Period, uint16 DutyCycle)

Function	void Pwm_SetPeriodAndDuty (Pwm_ChannelType ChannelNumber, Pwm_PeriodType Period, uint16 DutyCycle)	
Description	Set PWM period value and duty cycle.	
Diagram	N/A	
Parameters	Parameter	Description
	ChannelNumber	Input PWM channel number.
	Period	Input PWM period value.
	DutyCycle	Input PWM duty cycle value, the range is from 0x0000 to 0x8000.
Returns	N/A	

2.7.1.22 Std_ReturnType Pwm_SetPowerState (Pwm_PowerStateRequestResultType * Result)

Function	Std_ReturnType Pwm_SetPowerState (Pwm_PowerStateRequestResultType * Result)	
Description	PWM set power state.	
Diagram	N/A	
Parameters	Parameter	Description
	Result	Set power state result.
Returns	Call function result	

2.7.1.23 void Pwm_UnMaskOutputs (Pwm_ChannelType ChannelNumber)

Function	void Pwm_UnMaskOutputs (Pwm_ChannelType ChannelNumber)	
Description	PWM clear mask output, it will disable the forced channel.	
Diagram	N/A	
Parameters	Parameter	Description

	ChannelNumber	Input PWM channel number.
Returns	N/A	

2.8 Hardware Functions

2.8.1 Functions in Pwm_Hw.h.

2.8.1.1 void Pwm_Hw_Delnit (const Pwm_ConfigType * pPwmConfig)

Function	void Pwm_Hw_Delnit (const Pwm_ConfigType * pPwmConfig)	
Description	PWM high level De-initialization function	
Diagram	N/A	
Parameters	Parameter	Description
	pPwmConfig	PWM module configuration pointer
Returns	N/A	
Referenced By	void Pwm_Delnit(void)	

2.8.1.2 void Pwm_Hw_DisableNotification(const Pwm_ChannelConfigType * pPwmChannelConfig)

Function	void Pwm_Hw_DisableNotification(const Pwm_ChannelConfigType * pPwmChannelConfig)	
Description	PWM high level disable notification function.	
Diagram	 <pre> graph LR A[Pwm_Hw_DisableNotification] --> B[Pwm_Ftu_DisableNotification] </pre>	
Parameters	Parameter	Description
	pPwmChannelConfig	PWM channel configuration pointer
Returns	N/A	
Referenced By	void Pwm_DisableNotification(Pwm_ChannelType ChannelNumber)	

2.8.1.3 void Pwm_Hw_DisableTriggerOut (const Pwm_ChannelConfigType* pPwmChannelConfig, boolean bReloadTrig)

Function	void Pwm_Hw_DisableTriggerOut (const Pwm_ChannelConfigType* pPwmChannelConfig, boolean bReloadTrig)	
Description	PWM high level disable trigger out function.	
Diagram	 <pre> graph LR A[Pwm_Hw_DisableTriggerOut] --> B[Pwm_Ftu_DisableTriggerOut] </pre>	
Parameters	Parameter	Description
	pPwmChannelConfig	PWM channel configuration pointer
	bReloadTrig	Disable PWM channel reload trigger out.
Returns	N/A	
Referenced By	void Pwm_DisableTriggerOut(Pwm_ChannelType ChannelNumber, boolean bReloadTrig)	

2.8.1.4 void Pwm_Hw_EnableNotification (Pwm_EdgeNotificationType eNotification, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)

Function	void Pwm_Hw_EnableNotification (Pwm_EdgeNotificationType eNotification, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	
Description	PWM high level enable notification function.	

Diagram	 <pre> graph LR A[Pwm_Hw_EnableNotification] --> B[Pwm_Ftu_EnableNotification] </pre>	
Parameters	Parameter	Description
	eNotification	PWM edge detected type to generate notification.
	u8Channel	PWM channel number
	pPwmConfig	PWM module configuration pointer
Returns	N/A	
Referenced By	void Pwm_EnableNotification(Pwm_ChannelType ChannelNumber, Pwm_EdgeNotificationType Notification)	

2.8.1.5 void Pwm_HW_ConfigGTB(const uint32 u32Group, boolean benable)

Function	void Pwm_HW_ConfigGTB(const uint32 u32Group, boolean benable)	
Description	Config FTU global time base function	
Diagram		
Parameters	Parameter	Description
	u32Group	FTU global time base group number
	benable	enable/disable FTU global time base function
Returns	N/A	
Referenced By	void Pwm_StartGlobalTime(const uint32 u32Group) void Pwm_StopGlobalTime(const uint32 u32Group)	

2.8.1.6 void Pwm_Hw_EnableTriggerOut (const Pwm_ChannelConfigType* pPwmChannelConfig, boolean bReloadTrig)

Function	void Pwm_Hw_EnableTriggerOut (const Pwm_ChannelConfigType* pPwmChannelConfig, boolean bReloadTrig)	
Description	PWM high level enable trigger out function.	
Diagram	 <pre> graph LR A[Pwm_Hw_EnableTriggerOut] --> B[Pwm_Ftu_EnableTriggerOut] </pre>	
Parameters	Parameter	Description
	pPwmChannelConfig	PWM channel configuration pointer
	bReloadTrig	Enable PWM channel reload trigger out.
Returns	N/A	
Referenced By	void Pwm_EnableTriggerOut(Pwm_ChannelType ChannelNumber, boolean bReloadTrig)	

2.8.1.7 uint16 Pwm_Hw_GetChannelState (Pwm_ChannelType u8Chnnel, Pwm_InstanceType eInstance)

Function	uint16 Pwm_Hw_GetChannelState (Pwm_ChannelType u8Chnnel, Pwm_InstanceType eInstance)	
Description	PWM high level get channel state.	
Diagram	N/A	
Parameters	Parameter	Description
	u8Channel	PWM channel number
	eInstance	PWM hardware instance
Returns	channel state	
Referenced By	uint16 Pwm_GetChannelState(Pwm_ChannelType ChannelNumber)	

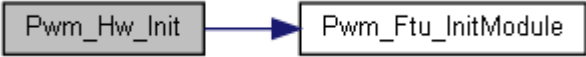
2.8.1.8 uint32 Pwm_Hw_GetMaxPeriodValue (void)

Function	uint32 Pwm_Hw_GetMaxPeriodValue (void)
Description	PWM high level get max period value.
Diagram	
Parameters	N/A
Returns	Max period value
Referenced By	void Pwm_SetPeriodAndDuty(Pwm_ChannelType ChannelNumber, Pwm_PeriodType Period, uint16 DutyCycle);

2.8.1.9 Pwm_OutputStateType Pwm_Hw_GetOutputState (const Pwm_ChannelConfigType* pPwmChannelConfig)

Function	Pwm_OutputStateType Pwm_Hw_GetOutputState (const Pwm_ChannelConfigType* pPwmChannelConfig)				
Description	PWM high level get output state function.				
Diagram					
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>pPwmChannelConfig</td><td>PWM channel configuration pointer</td></tr> </tbody> </table>	Parameter	Description	pPwmChannelConfig	PWM channel configuration pointer
Parameter	Description				
pPwmChannelConfig	PWM channel configuration pointer				
Returns	PWM channel output state				
Referenced By	Pwm_OutputStateType Pwm_GetOutputState(Pwm_ChannelType ChannelNumber)				

2.8.1.10 void Pwm_Hw_Init (const Pwm_ConfigType * pPwmConfig)

Function	void Pwm_Hw_Init (const Pwm_ConfigType * pPwmConfig)				
Description	PWM module high level initialization function.				
Diagram					
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>pPwmConfig</td><td>PWM module configuration pointer</td></tr> </tbody> </table>	Parameter	Description	pPwmConfig	PWM module configuration pointer
Parameter	Description				
pPwmConfig	PWM module configuration pointer				
Returns	N/A				
Referenced By	void Pwm_Init(const Pwm_ConfigType *ConfigPtr)				

2.8.1.11 void Pwm_Hw_MaskOutputs (const Pwm_ChannelConfigType* pPwmChannelConfig)

Function	void Pwm_Hw_MaskOutputs (const Pwm_ChannelConfigType* pPwmChannelConfig)				
Description	PWM high level mask output.				
Diagram					
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>pPwmChannelConfig</td><td>PWM channel configuration pointer</td></tr> </tbody> </table>	Parameter	Description	pPwmChannelConfig	PWM channel configuration pointer
Parameter	Description				
pPwmChannelConfig	PWM channel configuration pointer				
Returns	N/A				
Referenced By	void Pwm_MaskOutputs(Pwm_ChannelType ChannelNumber)				

2.8.1.12 void Pwm_Hw_SetDutyCycle (uint16 u16DutyCycle, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)

Function	void Pwm_Hw_SetDutyCycle (uint16 u16DutyCycle, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	
Description	PWM high level set duty cycle function.	
Diagram	N/A	
Parameters	Parameter	Description
	u16DutyCycle	PWM duty cycle value.
	u8Chnnel	PWM channel number
	pPwmConfig	PWM module configuration pointer
Returns	N/A	
Referenced By	void Pwm_SetDutyCycle(Pwm_ChannelType ChannelNumber, uint16 DutyCycle)	

2.8.1.13 void Pwm_Hw_SetOutputToldle (Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)

Function	void Pwm_Hw_SetOutputToldle (Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	
Description	PWM high level set output to idle state function.	
Diagram	N/A	
Parameters	Parameter	Description
	u8Chnnel	PWM channel number
	pPwmConfig	PWM module configuration pointer
Returns	N/A	
Referenced By	void Pwm_SetOutputToldle(Pwm_ChannelType ChannelNumber)	

2.8.1.14 void Pwm_Hw_SetPeriodAndDuty (Pwm_PeriodType u16Period, uint16 u16DutyCycle, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)

Function	void Pwm_Hw_SetPeriodAndDuty (Pwm_PeriodType u16Period, uint16 u16DutyCycle, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	
Description	PWM high level set period and duty cycle function.	
Diagram		
Parameters	Parameter	Description
	u16Period	PWM period value.
	u16DutyCycle	PWM duty cycle value.
	u8Chnnel	PWM channel number
	pPwmConfig	PWM module configuration pointer
Returns	N/A	
Referenced By	void Pwm_SetPeriodAndDuty(Pwm_ChannelType ChannelNumber, Pwm_PeriodType Period, uint16 DutyCycle)	

2.8.1.15 void Pwm_Hw_SetPowerState (const Pwm_ConfigType *pPwmConfig, Pwm_PowerStateType ePowerState)

Function	void Pwm_Hw_SetPowerState (const Pwm_ConfigType *pPwmConfig, Pwm_PowerStateType ePowerState)	
Description	PWM set power state.	

Diagram		
Parameters	Parameter	Description
	pPwmConfig	PWM module configuration pointer
	ePowerState	PWM power state enumeration.
Returns	N/A	
Referenced By	Std_ReturnType Pwm_SetPowerState(Pwm_PowerStateRequestResultType *Result)	

2.8.1.16 void Pwm_Hw_UnMaskOutputs (const Pwm_ChannelConfigType* pPwmChannelConfig)

Function	void Pwm_Hw_UnMaskOutputs (const Pwm_ChannelConfigType* pPwmChannelConfig)	
Description	PWM high level clear mask output. It will disable the forced channel.	
Diagram	N/A	
Parameters	Parameter	Description
	pPwmChannelConfig	PWM channel configuration pointer
Returns	N/A	
Referenced By	void Pwm_UnMaskOutputs(Pwm_ChannelType ChannelNumber)	

2.8.1.17 Std_ReturnType Pwm_Hw_ValidateIdleState (const Pwm_ConfigType *pPwmConfig)

Function	Std_ReturnType Pwm_Hw_ValidateIdleState (const Pwm_ConfigType *pPwmConfig)	
Description	PWM validate idle state.	
Diagram		
Parameters	Parameter	Description
	pPwmConfig	PWM module configuration pointer
Returns	PWM function return type.	
Referenced By	Std_ReturnType Pwm_SetPowerState(Pwm_PowerStateRequestResultType *Result)	

2.8.1.18 Std_ReturnType Pwm_Hw_ValidateNotification (Pwm_ChannelType u8Channel , const Pwm_ConfigType *pPwmConfig, Pwm_EdgeNotificationType eNotification)

Function	Std_ReturnType Pwm_Hw_ValidateNotification (Pwm_ChannelType u8Channel , const Pwm_ConfigType *pPwmConfig, Pwm_EdgeNotificationType eNotification)	
Description	PWM validate edge detected type to generate notification.	
Diagram		
Parameters	Parameter	Description
	u8Chnnel	PWM channel number
	pPwmConfig	PWM module configuration pointer
	eNotification	PWM edge detected type to generate notification.
Returns	PWM function return type	
Referenced By	void Pwm_EnableNotification(Pwm_ChannelType ChannelNumber, Pwm_EdgeNotificationType Notification)	

2.8.2 Functions in Pwm_Ftu.h

2.8.2.1 void Pwm_Ftu_DelnitChannel (const Pwm_ChannelConfigType* pFtuChannel)

Function	void Pwm_Ftu_DelnitChannel (const Pwm_ChannelConfigType* pFtuChannel)	
Description	PWM FTU low level De-initialization function.	
Diagram	N/A	
Parameters	Parameter	Description
	pFtuChannel	FTU channel configuration
Returns	N/A	
Referenced By	void Pwm_Hw_Delnit (const Pwm_ConfigType * pPwmConfig)	

2.8.2.2 void Pwm_Ftu_DelnitModule (const Pwm_Ftu_InstanceType eInstance)

Function	void Pwm_Ftu_DelnitModule (const Pwm_Ftu_InstanceType eInstance)	
Description	PWM low level common De-initialization function.	
Diagram	N/A	
Parameters	Parameter	Description
	eInstance	FTU instance
Returns	N/A	
Referenced By	void Pwm_Hw_Delnit (const Pwm_ConfigType * pPwmConfig)	

2.8.2.3 void Pwm_Ftu_DisableNotification (const Pwm_ChannelConfigType * pConfigChannel)

Function	Void Pwm_Ftu_DisableNotification (const Pwm_ChannelConfigType * pConfigChannel)	
Description	PWM FTU low level disable notification function.	
Diagram	N/A	
Parameters	Parameter	Description
	pConfigChannel	FTU channel configuration
Returns	N/A	
Referenced By	void Pwm_Hw_DisableNotification(const Pwm_ChannelConfigType * pPwmChannelConfig)	

2.8.2.4 void Pwm_Ftu_DisableTriggerOut (Pwm_InstanceType eFtuModule, uint8 u8FtuChannel, boolean bReloadTrig)

Function	void Pwm_Ftu_DisableTriggerOut (Pwm_InstanceType eFtuModule, uint8 u8FtuChannel, boolean bReloadTrig)	
Description	PWM FTU low level disable trigger out function.	
Diagram	N/A	
Parameters	Parameter	Description
	eFtuModule	FTU module.
	u8FtuChannel	FTU channel.
	bReloadTrig	FTU reload trigger enable or not.
Returns	N/A	
Referenced By	void Pwm_Hw_DisableTriggerOut (const Pwm_ChannelConfigType* pPwmChannelConfig, boolean bReloadTrig)	

2.8.2.5 void Pwm_Ftu_EnableNotification (uint16 u16DutyCycle, Pwm_EdgeNotificationType eNotification, Pwm_Ftu_AlignmentType eAlignment, const Pwm_ChannelConfigType * pConfigChannel)

Function	void Pwm_Ftu_EnableNotification (uint16 u16DutyCycle, Pwm_EdgeNotificationType eNotification, Pwm_Ftu_AlignmentType eAlignment, const Pwm_ChannelConfigType * pConfigChannel)	
Description	PWM FTU low level enable notification.	
Diagram	N/A	
Parameters	Parameter	Description
	u16DutyCycle	channel duty cycle
	eNotification	PWM edge detected to generate notification.
	eAlignment	PWM alignment
	pConfigChannel	FTU channel configuration
Returns	N/A	
Referenced By	void Pwm_Hw_EnableNotification (Pwm_EdgeNotificationType eNotification, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	

2.8.2.6 void Pwm_Ftu_EnableTriggerOut (Pwm_InstanceType eFtuModule, uint8 u8FtuChannel, boolean bReloadTrig)

Function	void Pwm_Ftu_EnableTriggerOut (Pwm_InstanceType eFtuModule, uint8 u8FtuChannel, boolean bReloadTrig)	
Description	PWM FTU low level enable trigger out function.	
Diagram	N/A	
Parameters	Parameter	Description
	eFtuModule	FTU module.
	u8FtuChannel	FTU channel.
	bReloadTrig	FTU reload trigger enable or not.
Returns	N/A	
Referenced By	void Pwm_Hw_EnableTriggerOut (const Pwm_ChannelConfigType* pPwmChannelConfig, boolean bReloadTrig)	

2.8.2.7 void Pwm_Ftu_ConfigGTB(const uint32 u32Group,boolean benable)

Function	void Pwm_Ftu_ConfigGTB(const uint32 u32Group,boolean benable)	
Description	Config the FTU GTB Function	
Diagram	N/A	
Parameters	Parameter	Description
	u8Group	FTU GTB Group number
	benable	enable/disable FTU GTB function.
Returns	N/A	
Referenced By	void Pwm_HW_ConfigGTB(const uint32 u32Group, boolean benable)	

2.8.2.8 uint32 Pwm_Ftu_GetMaxPeriodValue (void)

Function	uint32 Pwm_Ftu_GetMaxPeriodValue (void)
Description	Get max period value about FTU instance
Diagram	N/A

Parameters	N/A
Returns	Max period value
Referenced By	uint32 Pwm_Hw_GetMaxPeriodValue (void)

2.8.2.9 Pwm_OutputStateType Pwm_Ftu_GetOutputState (const Pwm_ChannelConfigType *pConfigChannel)

Function	Pwm_OutputStateType Pwm_Ftu_GetOutputState (const Pwm_ChannelConfigType *pConfigChannel)	
Description	PWM FTU low level get output state.	
Diagram	N/A	
Parameters	Parameter	Description
	pConfigChannel	FTU channel configuration
Returns	PWM FTU channel output state	
Referenced By	Pwm_OutputStateType Pwm_Hw_GetOutputState (const Pwm_ChannelConfigType* pPwmChannelConfig)	

2.8.2.10 void Pwm_Ftu_InitChannel (const Pwm_ChannelConfigType *pFtuChannel, Pwm_Ftu_AlignmentType eAlingment)

Function	void Pwm_Ftu_InitChannel (const Pwm_ChannelConfigType *pFtuChannel, Pwm_Ftu_AlignmentType eAlingment)	
Description	PWM low level initialization function.	
Diagram	N/A	
Parameters	Parameter	Description
	pConfigChannel	FTU channel configuration
	eAlingment	FTU channel alignment
Returns	N/A	
Referenced By	void Pwm_Init(const Pwm_ConfigType *ConfigPtr)	

2.8.2.11 void Pwm_Ftu_InitModule (const Pwm_Ftu_ModuleConfigType *pConfigModule)

Function	void Pwm_Ftu_InitModule (const Pwm_Ftu_ModuleConfigType *pConfigModule)	
Description	PWM low level initialize static parameters.	
Diagram	N/A	
Parameters	Parameter	Description
	pConfigModule	FTU module configuration
Returns	N/A	
Referenced By	void Pwm_Init(const Pwm_ConfigType *ConfigPtr)	

2.8.2.12 void Pwm_Ftu_MaskOutputs (Pwm_InstanceType eModuleId, uint8 u8ChannelMask)

Function	void Pwm_Ftu_MaskOutputs (Pwm_InstanceType eModuleId, uint8 u8ChannelMask)	
Description	PWM FTU low level mask output function.	
Diagram	N/A	
Parameters	Parameter	Description
	eModuleId	FTU module.

	u8ChannelMask	FTU channel mask value.
Returns	N/A	
Referenced By	void Pwm_Hw_MaskOutputs (const Pwm_ChannelConfigType* pPwmChannelConfig, uint8 u8ChannelMask)	

2.8.2.13 void Pwm_Ftu_SetDutyCycle (uint16 u16DutyCycle, const Pwm_ChannelConfigType *pFtuChannel, const Pwm_Ftu_ModuleConfigType *pModule)

Function	void Pwm_Ftu_SetDutyCycle (uint16 u16DutyCycle, const Pwm_ChannelConfigType *pFtuChannel, const Pwm_Ftu_ModuleConfigType *pModule)	
Description	PWM low level set duty cycle value.	
Diagram	N/A	
Parameters	Parameter	Description
	u16DutyCycle	PWM duty cycle value.
	pConfigChannel	FTU channel configuration
	pModule	FTU module configuration
Returns	N/A	
Referenced By	void Pwm_Hw_SetDutyCycle (uint16 u16DutyCycle, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	

2.8.2.14 void Pwm_Ftu_SetOutputToldle (const Pwm_ChannelConfigType *pConfigChannel)

Function	void Pwm_Ftu_SetOutputToldle (const Pwm_ChannelConfigType *pConfigChannel)	
Description	PWM FTU low level set output to idle function.	
Diagram	N/A	
Parameters	Parameter	Description
	pConfigChannel	FTU channel configuration
Returns	N/A	
Referenced By	void Pwm_Hw_SetOutputToldle (Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	

2.8.2.15 void Pwm_Ftu_SetPeriodAndDuty (Pwm_PeriodType u16Period, uint16 u16DutyCycle, const Pwm_ChannelConfigType *pFtuChannel, const Pwm_Ftu_ModuleConfigType *pModule)

Function	void Pwm_Ftu_SetPeriodAndDuty (Pwm_PeriodType u16Period, uint16 u16DutyCycle, const Pwm_ChannelConfigType *pFtuChannel, const Pwm_Ftu_ModuleConfigType *pModule)	
Description	PWM FTU low level set period and duty cycle function.	
Diagram	N/A	
Parameters	Parameter	Description
	u16Period	PWM FTU period value.
	u16DutyCycle	PWM FTU duty cycle value.
	pFtuChannel	FTU channel configuration
	pModule	FTU module configuration
Returns	N/A	
Referenced By	void Pwm_SetPeriodAndDuty(Pwm_ChannelType ChannelNumber, Pwm_PeriodType Period, uint16 DutyCycle)	

2.8.2.16 void Pwm_Ftu_SetPowerState (const Pwm_Ftu_ModuleConfigType *pModule, Pwm_PowerStateType ePowerState)

Function	void Pwm_Ftu_SetPowerState (const Pwm_Ftu_ModuleConfigType *pModule, Pwm_PowerStateType ePowerState)	
Description	PWM FTU low level set power state function.	
Diagram	N/A	
Parameters	Parameter	Description
	pModule	FTU module configuration
	ePowerState	PWM power state.
Returns	N/A	
Referenced By	void Pwm_Hw_SetPowerState (const Pwm_ConfigType *pPwmConfig, Pwm_PowerStateType ePowerState)	

2.8.2.17 void Pwm_Ftu_UnMaskOutputs (Pwm_InstanceType eModuleId, uint8 u8ChannelMask)

Function	void Pwm_Ftu_UnMaskOutputs (Pwm_InstanceType eModuleId, uint8 u8ChannelMask)	
Description	PWM low level clear mask output, it will disable the forced channel.	
Diagram	N/A	
Parameters	Parameter	Description
	eModuleId	FTU module.
	u8ChannelMask	FTU channel mask value .
Returns	N/A	
Referenced By	void Pwm_Hw_UnMaskOutputs (const Pwm_ChannelConfigType* pPwmChannelConfig, uint8 u8ChannelMask)	

2.8.2.18 Std_ReturnType Pwm_Ftu_ValidateNotification (Pwm_Ftu_AlignmentType eAlignment, Pwm_EdgeNotificationType eNotification)

Function	Std_ReturnType Pwm_Ftu_ValidateNotification (Pwm_Ftu_AlignmentType eAlignment, Pwm_EdgeNotificationType eNotification)	
Description	PWM FTU low level validate notification function.	
Diagram	N/A	
Parameters	Parameter	Description
	eAlignment	FTU align type.
	eNotification	FTU edge detected type to generate notification.
Returns	PWM FTU function return type	
Referenced By	Std_ReturnType Pwm_Hw_ValidateNotification (Pwm_ChannelType u8Channel , const Pwm_ConfigType *pPwmConfig, Pwm_EdgeNotificationType eNotification)	

2.8.2.19 void Pwm_Ftu_ClearOutputForce(const Pwm_ChannelConfigType *pConfigChannel)

Function	void Pwm_Ftu_ClearOutputForce(const Pwm_ChannelConfigType *pConfigChannel)
Description	PWM FTU low level clear output force
Diagram	N/A

Parameters	Parameter	Description
	pConfigChanne	FTU channel configuration
Returns	N/A	
Referenced By	void Pwm_Hw_SetPeriodAndDuty (Pwm_PeriodType u16Period, uint16 u16DutyCycle, Pwm_ChannelType u8Channel, const Pwm_ConfigType * pPwmConfig)	

2.9 Peripheral Functions

2.9.1 Functions

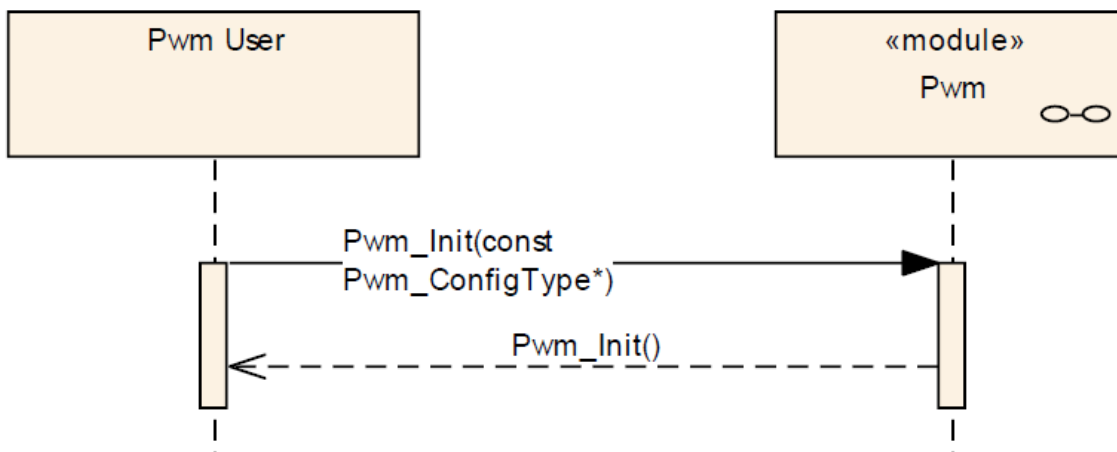
FTU peripheral functions defined in Ftu_RegOps.h at Common module.

2.10 API Sequence Diagram

2.10.1 Initialization

PWM Driver Initialization

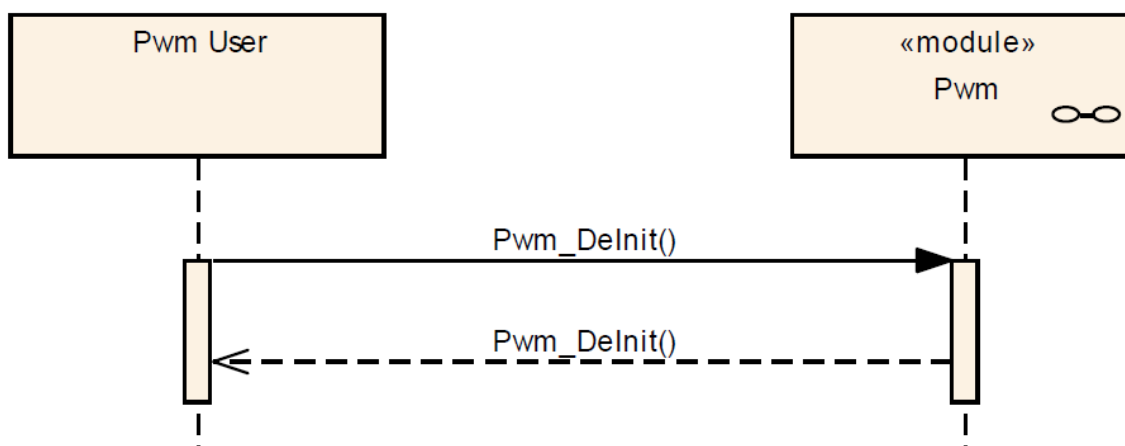
The PWM output signals are either in low state, in high state or in modulation state depending on the configuration parameters. If configured, no notification occurs until the first call of Pwm_EnableNotification.



2.10.2 De-initialization

PWM Driver De-Initialization

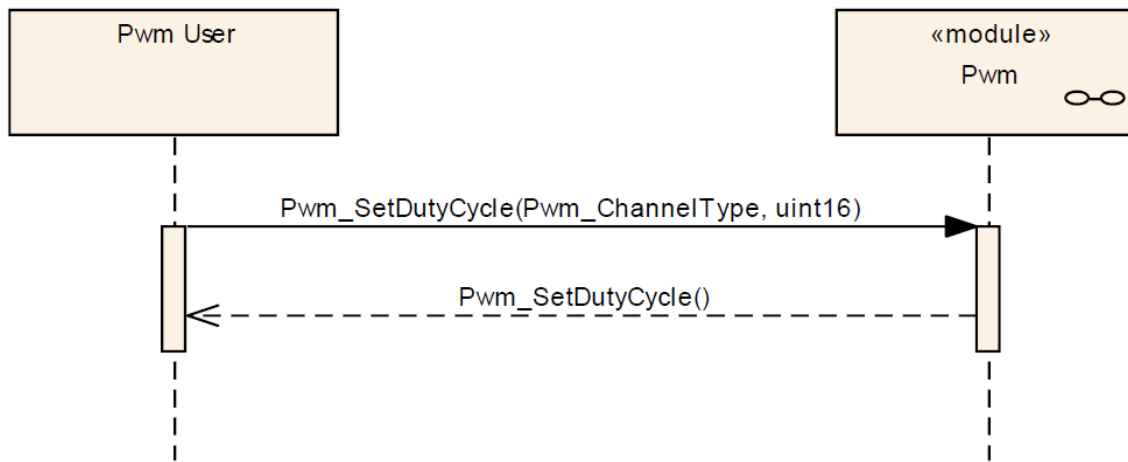
The PWM output channels are in the state defined by configuration.



2.10.3 Setting the Duty Cycle

Set PWM Driver Set Duty Cycle

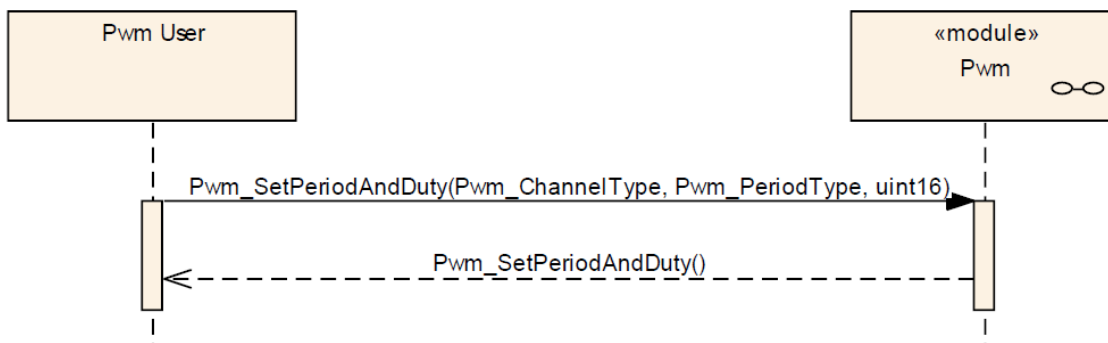
The PWM duty cycle will be changed either at the end of the current period if supported or directly if not supported by the implementation.



2.10.4 Setting the Period and the Duty

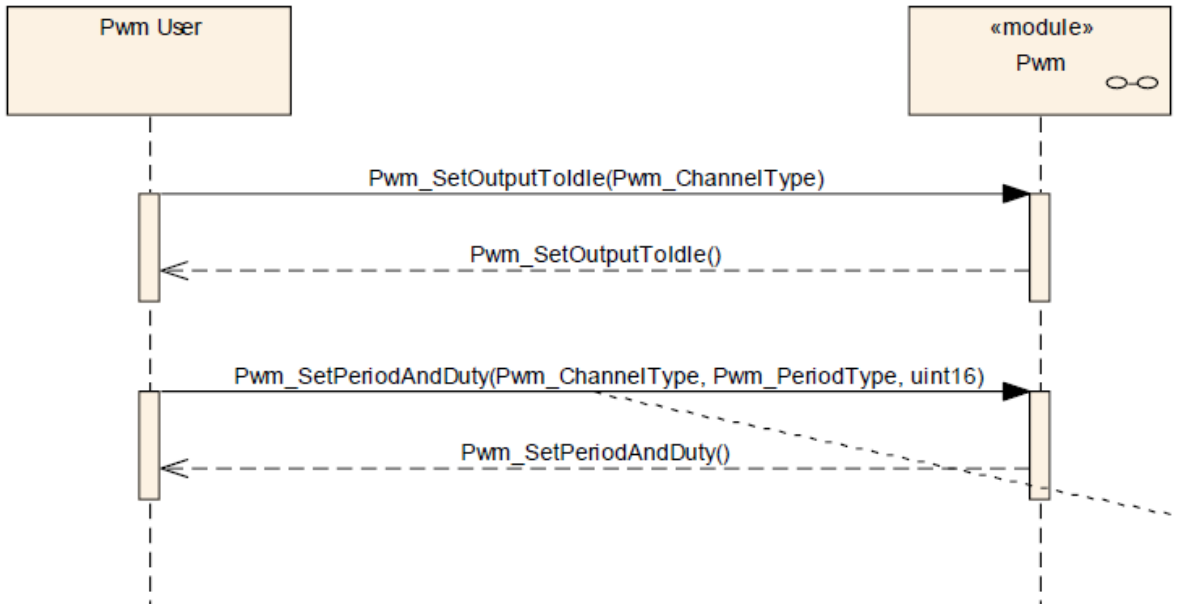
Set PWM signal period

The PWM period is changed at the end of the current period if configured.



2.10.5 Setting the PWM Output to Idle

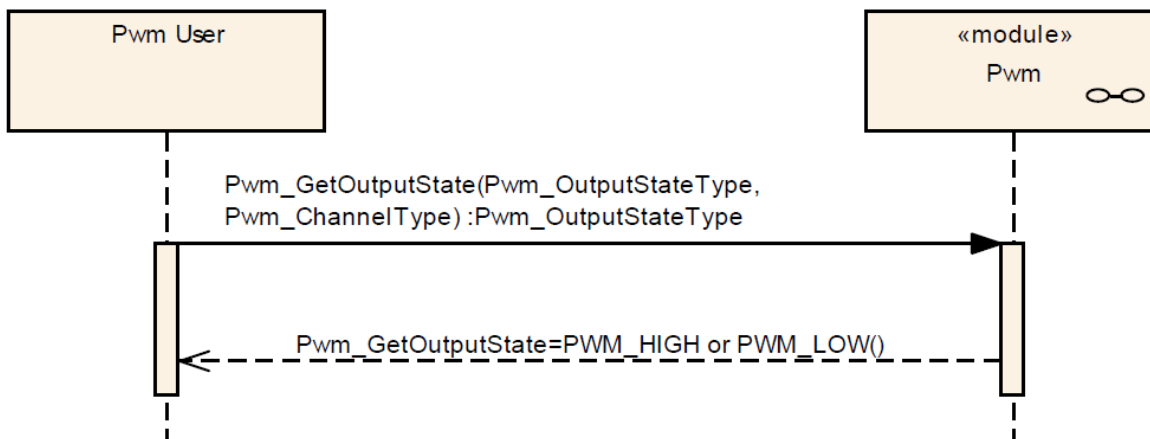
Set PWM signal



2.10.6 Getting the PWM Output State

Getting the PWM Output State

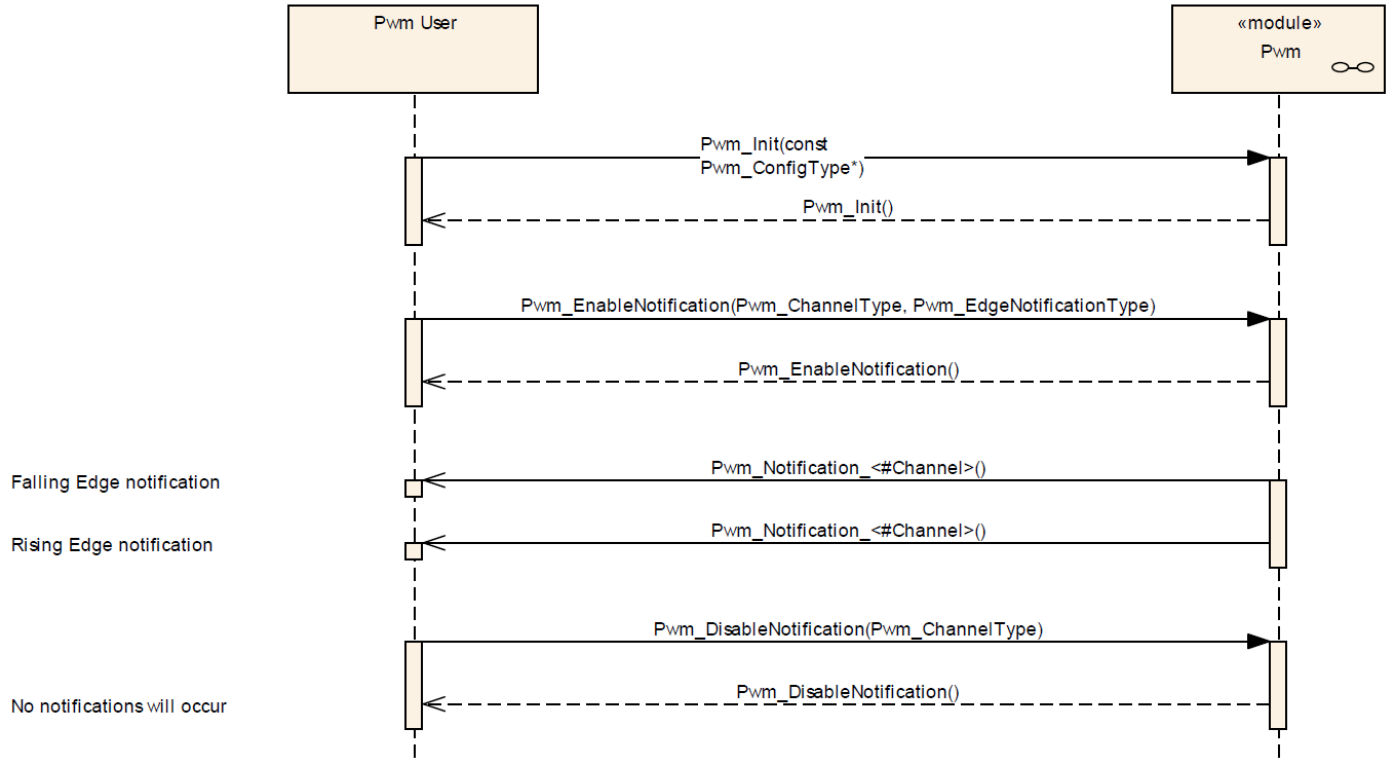
The PWM channel state is read.



2.10.7 Using the PWM Notifications

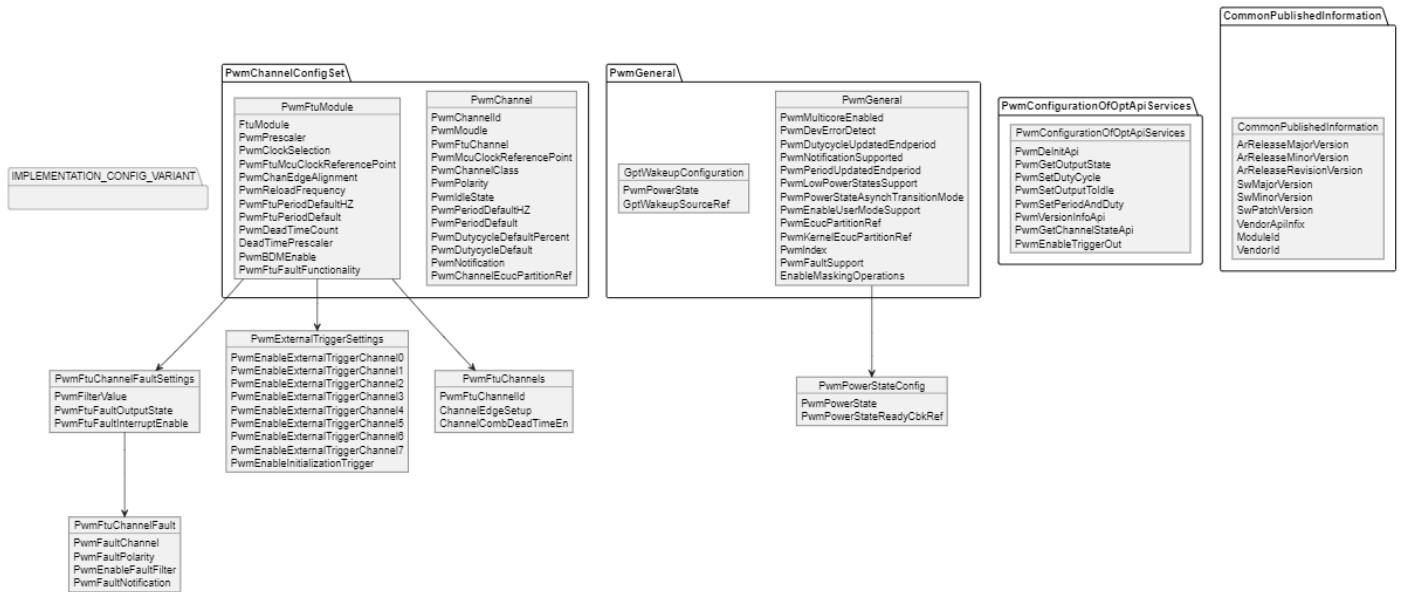
Using PWM Enable/Disable notification

The PWM channel is autostarted (Modulation starts during the call of `Pwm_Init()`). No notifications occur until the first call of `Pwm_EnableNotification(...)`. `Pwm_Notification<#Channel>` represents the configured callback function for a channel.



Chapter 3 Tresos Configuration Items

3.1 Container Inclusion Relation



3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Variable	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 PwmConfigurationOfOptApiServices

Container	PwmConfigurationOfOptApiServices	
Description	PwmConfigurationOfOptApiServices container used for storing the configuration of all optional API's.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1 PwmDelInitApi

Variable	PwmDelInitApi	
Description	Adds / Removes the service Pwm_DelInit() from the code.	
Screenshot	PwmDelInitApi  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.2 PwmGetOutputState

Variable	PwmGetOutputState	
Description	Adds / Removes the service Pwm_GetOutputState() from the code.	
Screenshot	PwmGetOutputState  ☐  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.3 PwmSetDutyCycle

Variable	PwmSetDutyCycle	
Description	Adds / Removes the service Pwm_SetDutyCycle() from the code.	
Screenshot	PwmSetDutyCycle  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.4 PwmSetOutputToldle

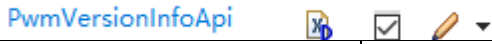
Variable	PwmSetOutputToldle	
Description	Adds / removes the service Pwm_SetOutputToldle() from the code.	
Screenshot	PwmSetOutputToldle  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.5 PwmSetPeriodAndDuty

Variable	PwmSetPeriodAndDuty	
Description	Adds / Removes the service Pwm_SetPeriodAndDuty() from the code.	
Screenshot	PwmSetPeriodAndDuty  ☒  	
Properties	Property	Value

	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.6 PwmVersionInfoApi

Variable	PwmVersionInfoApi	
Description	Switch to indicate that the Pwm_GetVersionInfo is supported.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

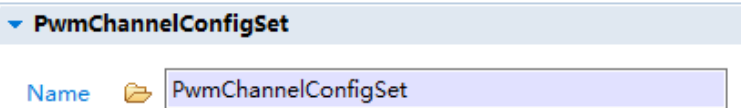
3.2.2.7 PwmGetChannelStateApi

Variable	PwmGetChannelStateApi	
Description	Switch to indicate that the Pwm_GetChannelState is supported.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

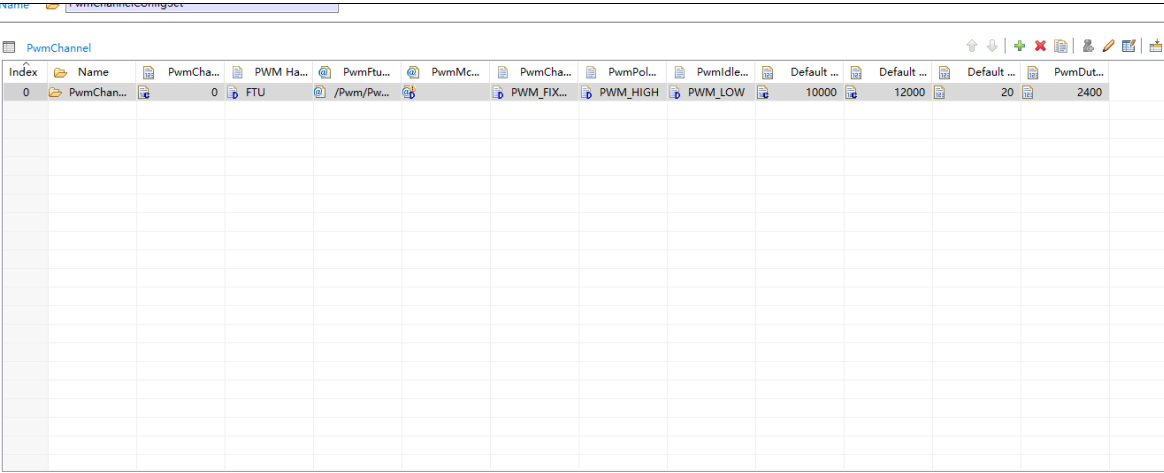
3.2.2.8 PwmEnableTriggerOut

Variable	PwmEnableTriggerOut	
Description	Adds / Removes the service Pwm_EnableTriggerOut()/Pwm_EnableTriggerOut() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

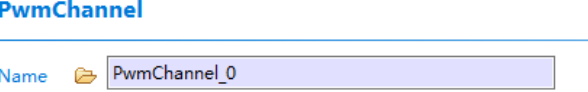
3.2.3 PwmChannelConfigSet

Container	PwmChannelConfigSet	
Description	This container contains the Channel configuration parameter of the PWM driver.	
Screenshot		
Properties	Property	Value
	Type	Container

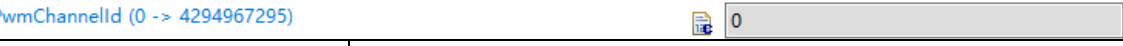
3.2.4 PwmChannel

Container	PwmChannel	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Map
	MIN	1

3.2.4.1 PwmChannel

Container	PwmChannel	
Description	Configuration of an individual PWM channel.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.4.1.1 PwmChannelId

Variable	PwmChannelId	
Description	Channel Id of the PWM channel. This value will be assigned to the symbolic name derived of the PwmChannel container short name.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	true
	Range	0~ 4294967295
	Editable	false

3.2.4.1.2 PwmMoudle

Variable	PwmMoudle	
Description	Hardware peripheral to be used for current PWM channel: FTU - FTU Hardware IP will be used. Please select a FTU configured channel from PwmFtuChannel	

	parameter below.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	FTU

3.2.4.1.3 PwmFtuChannel

Variable	PwmFtuChannel	
Description	Select the FTU channel on which the functionality of the current PWM channel will be implemented.	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	FLAGCHIP

3.2.4.1.4 PwmMcuClockReferencePoint

Variable	PwmMcuClockReferencePoint	
Description	Reference to the PWM clock source configuration which is set in the MCU driver configuration	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	AUTOSAR_ECUC
	Editable	false
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSettingConfig/McuClockReferencePoint

3.2.4.1.5 PwmChannelClass

Variable	PwmChannelClass	
Description	Class of PWM Channel	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	PWM_FIXED_PERIOD; PWM_VARIABLE_PERIOD; PWM_FIXED_PERIOD_SHIFTED

3.2.4.1.6 PwmPolarity

Variable	PwmPolarity
Description	Defines the starting polarity of each PWM channel.

Screenshot	 	
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	PWM_HIGH; PMW_LOW

3.2.4.1.7 PwmIdleState

Variable	PwmIdleState	
Description	This parameter represents the state of the Pin when the output is set to Idle.	
Screenshot	 	
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	PWM_HIGH; PMW_LOW

3.2.4.1.8 PwmPeriodDefaultHZ

Variable	PwmPeriodDefaultHZ	
Description	Default period HZ.	
Screenshot	 	
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.4.1.9 PwmPeriodDefault

Variable	PwmPeriodDefault	
Description	Default period value used at initialization.	
Screenshot	 	
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.4.1.10 PwmDutycycleDefault

Variable	PwmDutycycleDefault	
Description	Default value for duty cycle used for initialization.	
Screenshot	 	
Properties	Property	Value
	Type	Variable: INTEGER

	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	0~0x8000

3.2.4.1.11 PwmPhaseShiftValue

Variable	PwmPhaseShiftValue	
Description	The Pwm phase shift value is used to set the offset time of the PWM pulse signal	
Screenshot		
Properties	Property	Value
	Type	Variable: FLOAT
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.4.1.12 PwmNotification

Variable	PwmNotification	
Description	User callback function.	
Screenshot		
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Optional	true

3.2.4.1.13 PwmChannelEcucPartitionRef

Variable	PwmChannelEcucPartitionRef	
Description	Maps a Pwm channel to zero or multiple ECUC partitions to limit the access to this channel group. The ECUC partitions referenced are a subset of the ECUC partitions where the Pwm driver is mapped to. When users choose ENABLE multicore feature by checking PwmMulticoreEnabled option, at least one ECUC partition referenced from the ECUC module needs to be configured in this list.	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	AUTOSAR_ECUC

3.2.5 PwmFtuModule

Container	PwmFtuModule
Description	N/A

Screenshot

Properties

Property	Value
Type	Map
MIN	0

3.2.5.1 PwmFtuModule

Container	PwmFtuModule	
Description	Configuration of a FTU module available on the platform.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.5.1.1 FtuModule

Variable	FtuModule	
Description	Select one of the 4 sub-modules available on the current FTU module.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false

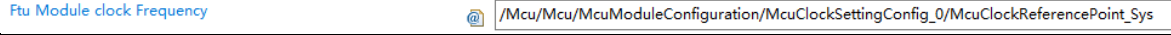
3.2.5.1.2 PwmPrescaler

Variable	PwmPrescaler	
Description	Autosar Requirements: Not specified.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	PRESC_1; PRESC_2; PRESC_4; PRESC_8; PRESC_16; PRESC_32; PRESC_64; PRESC_128

3.2.5.1.3 PwmClockSelection

Variable	PwmClockSelection	
Description	FTU Module clock selection.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	PWM_SYSTEM_CLOCK; PWM_EXTERNAL_CLOCK; PWM_NO_CLOCK

3.2.5.1.4 PwmFtuMcuClockReferencePoint

Variable	PwmFtuMcuClockReferencePoint	
Description	Reference to the FTU clock source configuration which is set in the MCU driver configuration.	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	AUTOSAR_ECUC
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSettingConfig/McuClockReferencePoint

3.2.5.1.5 PwmChanEdgeAlignment

Variable	PwmChanEdgeAlignment	
Description	This list will change the PWM generation mode for this FTU sub module.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	PWM_EDGE_ALIGNED; PWM_CENTER_ALIGNED

3.2.5.1.6 PwmReloadFrequency

Variable	PwmReloadFrequency	
Description	Define the number of enabled reload opportunities should happen until an enabled reload opportunity becomes a reload point.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	LDFQ_EACH1 ~ LDFQ_EACH32

3.2.5.1.7 PwmFtuPeriodDefaultHZ

Variable	PwmFtuPeriodDefaultHZ	
Description	Default FTU period HZ.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.8 PwmFtuPeriodDefault

Variable	PwmFtuPeriodDefault	
Description	Default period value used at initialization.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	0~65534

3.2.5.1.9 PwmDeadTimeCount

Variable	PwmDeadTimeCount	
Description	DeadTime Counter used to create a phase offset between the output of two consecutive FTU channels used complementary and combine modes.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0~1023
	Optional	true

3.2.5.1.10 DeadTimePrescaler

Variable	DeadTimePrescaler	
Description	DeadTime counter prescaler.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	PRESC_1; PRESC_4; PRESC_16

	Optional	true
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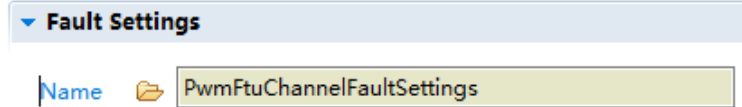
3.2.5.1.11 PwmBDMEnable

Variable	PwmBDMEnable	
Description	Selects the Debug mode.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	CNR_STOPED_FLAG_SET; CNT_STOPED_OUTPUTS_SAFE; CNT_STOPED_OUTPUTS_FROZEN; CNT_ON_OUTPUTS_ON

3.2.5.1.12 PwmFtuFaultFunctionality

Variable	PwmFtuFaultFunctionality	
Description	Selects the fault control mode.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	FLTCTRL_DISABLED; MANUAL_FOR_EVEN_CHANNELS; MANUAL_FOR_ALL_CHANNELS; AUTOMATIC_FOR_ALL_CHANNELS

3.2.5.1.13 PwmFtuChannelFaultSettings

Container	PwmFtuChannelFaultSettings	
Description	Global Fault configuration.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.5.1.14 PwmFilterPrescaler

Variable	PwmFilterPrescaler	
Description	Selects the filter prescaler for the FTU module.	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	FTU_FLT_DIV_1-- FTU_FLT_DIV_16

3.2.5.1.15 PwmFilterValue

Variable	PwmFilterValue	
Description	Value used for debouncing Fault inputs.	
Screenshot		
Properties	Property	Value
	Type	Variable:INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0~15

3.2.5.1.16 PwmFtuFaultOutputState

Variable	PwmFtuFaultOutputState	
Description	Specifies the fault state for the PWM channel output during fault conditions	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	SAFE_VALUE; TRISTATE

3.2.5.1.17 PwmFtuFaultInterruptEnable

Variable	PwmFtuFaultInterruptEnable	
Description	Whether to enable fault interrupt function.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.18 PwmFtuChannelFaultSettings

Container	PwmFtuChannelFaultSettings
Description	Whether to enable fault interrupt function.

Screenshot

PwmFtuChannelFaultSettings

Index

Name

Fault Inp...

Configur...

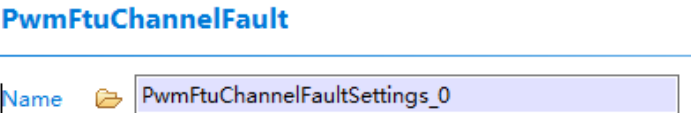
Enable F...

PwmFaul...

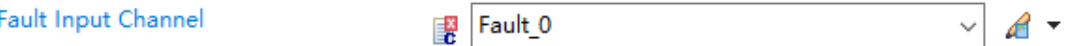
Properties

Property	Value
Type	MAP
MAX	4

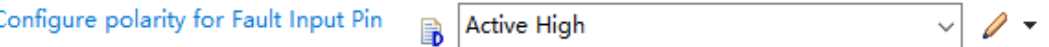
3.2.5.1.19 PwmFtuChannelFault

Container	PwmFtuChannelFault	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.5.1.20 PwmFaultChannel

Variable	PwmFaultChannel	
Description	Select fault input channel.	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.21 PwmFaultPolarity

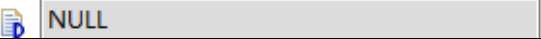
Variable	PwmFaultPolarity	
Description	Configure polarity for Fault Input Pin.	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false

	Range	Active Low; Active High
--	-------	-------------------------

3.2.5.1.22 PwmEnableFaultFilter

Variable	PwmEnableFaultFilter	
Description	Enables glitch filter for Fault Input pin.	
Screenshot	Enable Fault Input Glitch Filter 	
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

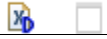
3.2.5.1.23 PwmFaultNotification

Variable	PwmFaultNotification	
Description	User callback function.	
Screenshot	PwmFaultNotification 	
Properties	Property	Value
	Type	Variable : FUNCTION-NAME
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Optional	false

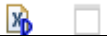
3.2.5.1.24 PwmExternalTriggerSettings

Container	PwmExternalTriggerSettings	
Description	Global Trigger configuration.	
Screenshot	N/A	
Properties	Property	Value
	Type	Container

3.2.5.1.25 PwmEnableExternalTriggerChannel0

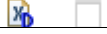
Variable	PwmEnableExternalTriggerChannel0	
Description	Enables the generation of the channel trigger on Trigger channel 0.	
Screenshot	Enable External Trigger Channel 0 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.26 PwmEnableExternalTriggerChannel1

Variable	PwmEnableExternalTriggerChannel1	
Description	Enables the generation of the channel trigger on Trigger channel 1.	
Screenshot	Enable External Trigger Channel 1 	

Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.27 PwmEnableExternalTriggerChannel2

Variable	PwmEnableExternalTriggerChannel2	
Description	Enables the generation of the channel trigger on Trigger channel 2.	
Screenshot	Enable External Trigger Channel 2 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.28 PwmEnableExternalTriggerChannel3

Variable	PwmEnableExternalTriggerChannel3	
Description	Enables the generation of the channel trigger on Trigger channel 3.	
Screenshot	Enable External Trigger Channel 3 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.29 PwmEnableExternalTriggerChannel4

Variable	PwmEnableExternalTriggerChannel4	
Description	Enables the generation of the channel trigger on Trigger channel 4.	
Screenshot	Enable External Trigger Channel 4 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.30 PwmEnableExternalTriggerChannel5

Variable	PwmEnableExternalTriggerChannel5	
Description	Enables the generation of the channel trigger on Trigger channel 5.	
Screenshot	Enable External Trigger Channel 5 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.31 PwmEnableExternalTriggerChannel6

Variable	PwmEnableExternalTriggerChannel6	
Description	Enables the generation of the channel trigger on Trigger channel 6.	
Screenshot	Enable External Trigger Channel 6 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

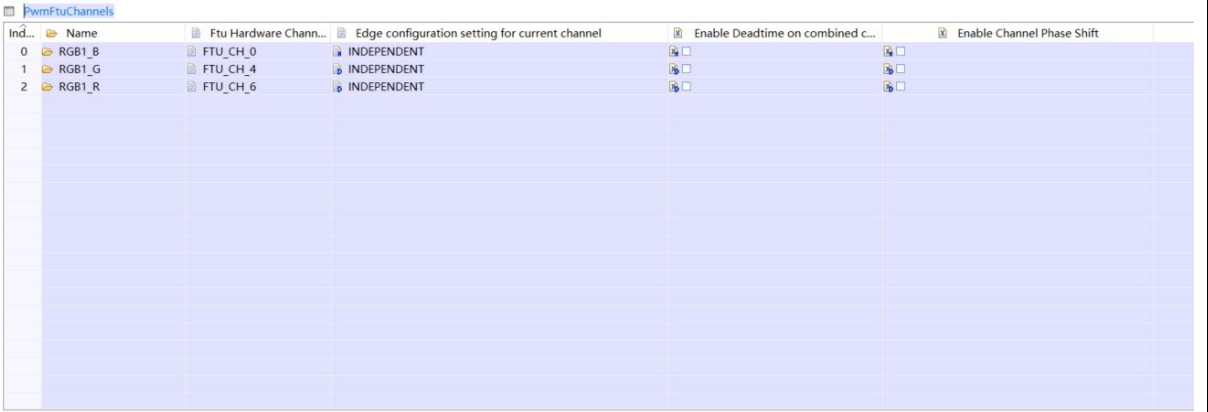
3.2.5.1.32 PwmEnableExternalTriggerChannel7

Variable	PwmEnableExternalTriggerChannel7	
Description	Enables the generation of the channel trigger on Trigger channel 7.	
Screenshot	Enable External Trigger Channel 7 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.33 PwmEnableInitializationTrigger

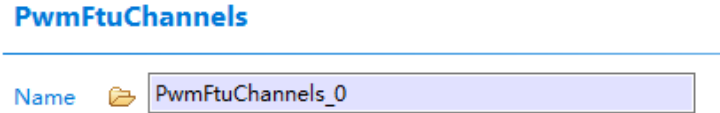
Variable	PwmEnableInitializationTrigger	
Description	Enables the generation of the trigger when the FTU counter is equal to the CNTIN register.	
Screenshot	Enable Reload Trigger 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.34 PwmFtuChannels

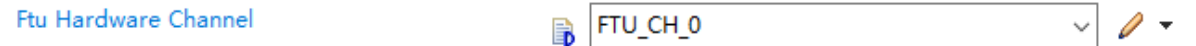
Container	PwmFtuChannels	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	MAP

	MIN	1
	MAX	8

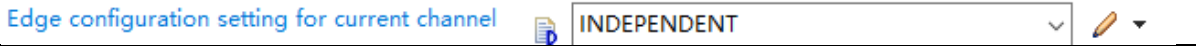
3.2.5.1.35 PwmFtuChannels

Container	PwmFtuChannels	
Description	Configuration of an individual PWM FTU channel.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.5.1.36 PwmFtuChannelId

Variable	PwmFtuChannelId	
Description	Channel Id of the Flex Timer channel of the current. This value will be assigned to the symbolic name derived of the PwmChannel container short name.	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	FTU_CH_0 ~ FTU_CH_7

3.2.5.1.37 ChannelEdgeSetup

Variable	ChannelEdgeSetup	
Description	Edge configuration setting for current channel	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	INDEPENDENT; COMBINED_SYNCED; COMBINED_COMPLEMENTARY

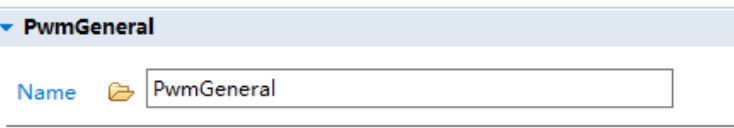
3.2.5.1.38 ChannelCombDeadTimeEn

Variable	ChannelCombDeadTimeEn	
Description	Switch for enabling / disabling DeadTime for COMBINED_COMPLEMENTARY channels.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.5.1.39 ChannelPhaseShiftEn

Variable	ChannelPhaseShiftTimeEn	
Description	Switch for enabling / disabling Phase Shift for COMBINED_SYNCED/COMBINED_COMPLEMENTARY channels.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6 PwmGeneral

Container	PwmGeneral	
Description	PwmGeneral container used for storing the general configuration of PWM.	
Screenshot		
Properties	Property	Value
	Type	Container

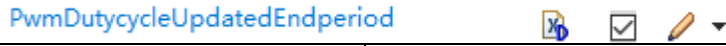
3.2.6.1 PwmMulticoreEnabled

Variable	PwmMulticoreEnabled	
Description	Switch to enable/disable multicore feature. When enabling this feature, at least one ECUC partition needs to be configured in the PwmChannelEcucPartitionRef container, and each Pwm channel in the PwmChannel container also needs to be configured with at least one ECUC partition.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.6.2 PwmDevErrorDetect

Variable	PwmDevErrorDetect	
Description	Switch for enabling / disabling the development error detection.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.6.3 PwmDutycycleUpdatedEndperiod

Variable	PwmDutycycleUpdatedEndperiod	
Description	Switch for enabling the update of the duty cycle parameter at the end of the current period.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.6.4 PwmNotificationSupported

Variable	PwmNotificationSupported	
Description	Switch to indicate that the notifications are supported.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

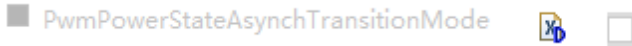
3.2.6.5 PwmPeriodUpdatedEndperiod

Variable	PwmPeriodUpdatedEndperiod	
Description	Switch for enabling the update of the period parameter at the end of the current period.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Readonly	true

3.2.6.6 PwmLowPowerStatesSupport

Variable	PwmLowPowerStatesSupport	
Description	If enabled, Hardware offer low power mode and Adds all power state management related APIs (PWM_SetPowerState, PWM_GetCurrentPowerState, PWM_GetTargetPowerState, PWM_PreparePowerState, PWM_Main_PowerTransitionManager), indicating if the HW offers low power state management.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Optional	true

3.2.6.7 PwmPowerStateAsynchTransitionMode

Variable	PwmPowerStateAsynchTransitionMode	
Description	Enables / disables support of the PWM Driver to the asynchronous power state transition.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Optional	true
	Editable	false

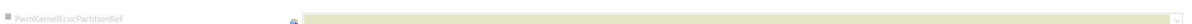
3.2.6.8 PwmEnableUserModeSupport

Variable	PwmEnableUserModeSupport	
Description	Enables / disables support of the PWM Driver to the asynchronous power state transition.	
Screenshot		
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Optional	true
	Editable	false

3.2.6.9 PwmEcucPartitionRef

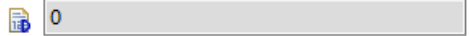
Variable	PwmEcucPartitionRef	
Description	Maps the Pwm driver to zero or multiple ECUC partitions to make the driver API available in the according partition. When users choose ENABLE multicore feature by checking PwmMulticoreEnabled option, At least one ECUC partition referenced from the ECUC module needs to be configured in this list.	
Screenshot		
Properties	Property	Value
	Type	Variable : REFERENCE
	Origin	AUTOSAR_ECUC
	Scope	ECU

3.2.6.10 PwmKernelEcucPartitionRef

Variable	PwmKernelEcucPartitionRef	
Description	Maps the Pwm kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the Pwm driver is mapped to. This feature is not supported and is rejected.	
Screenshot		

Properties	Property	Value
	Type	Variable : REFERENCE
	Origin	AUTOSAR_ECUC
	Optional	true
	Readonly	true

3.2.6.11 PwmIndex

Variable	PwmIndex	
Description	Specifies the Instance Id of this module instance. If only one instance is present it shall have the Id 0.	
Screenshot	PwmIndex (0 -> 4294967295) 	
Properties	Property	Value
	Type	Variable : INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Readonly	true
	Range	0~ 4294967295

3.2.6.12 PwmFaultSupport

Variable	PwmFaultSupport	
Description	This enables the fault functionality.	
Screenshot	Fault Support Enable 	
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6.13 EnableMaskingOperations

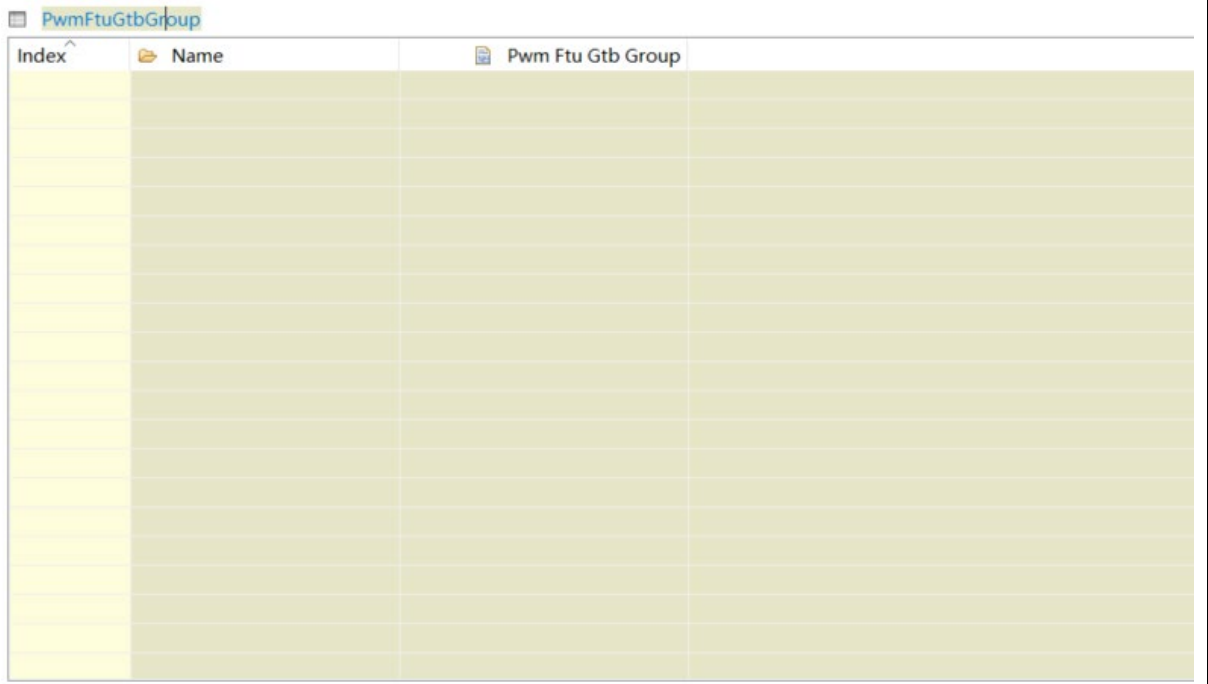
Variable	EnableMaskingOperations	
Description	Adds / Removes the service Pwm_MaskOutputs() and Pwm_UnMaskOutputs() from the code.	
Screenshot	EnableMaskingOperations 	
Properties	Property	Value
	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6.14 PwmGtbSupport

Variable	PwmGtbSupport	
Description	The FTU supports a global time base through the Global Time Base (GTB) feature. When this parameter is enabled, the ftu CNT can be control to count at the same time by calling Pwm_ConfigGTB().	
Screenshot	Pwm Global Time Base Support 	
Properties	Property	Value

	Type	Variable : BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6.15 PwmFtuGtbGroup

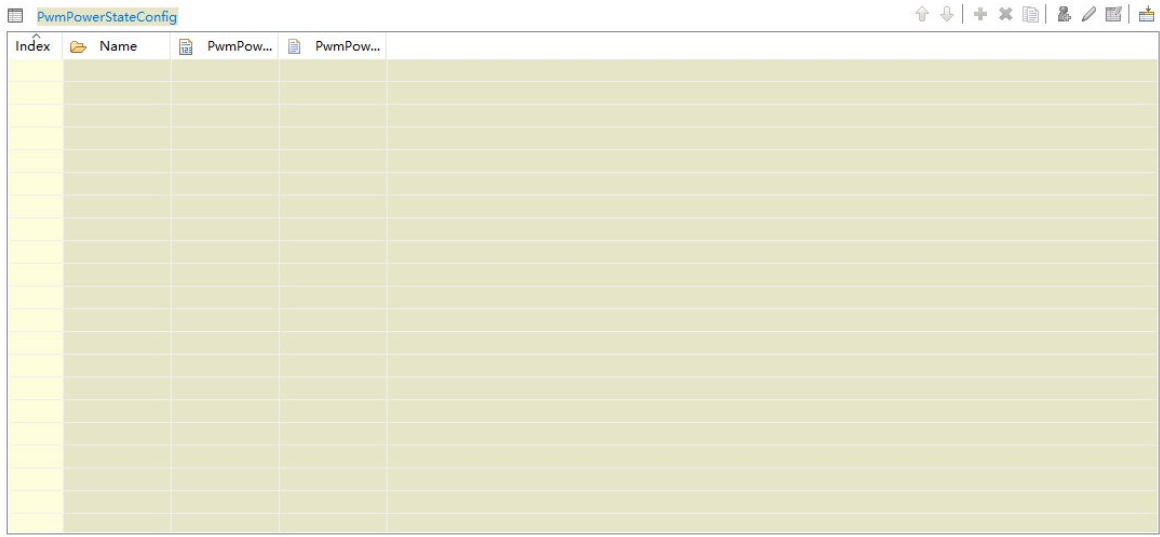
Container	PwmFtuGtbGroup	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	MAP

3.2.6.15.1 PwmFtuGroup

Variable	PwmFtuGroup	
Description	Set the Ftu Gtb Group.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6.16 PwmPowerStateConfig

Container	PwmPowerStateConfig	
Description	N/A	

Screenshot		
Properties	Property	Value
	Type	MAP

3.2.6.17 PwmPowerStateConfig

Container	PwmPowerStateConfig	
Description	N/A	
Screenshot	N/A	
Properties	Property	Value
	Type	Container

3.2.6.17.1 PwmPowerState

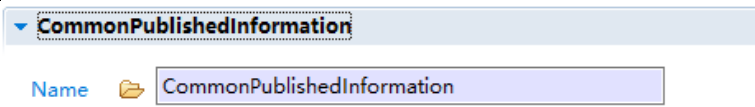
Variable	PwmPowerState	
Description	Each instance of this parameter describes a different power state supported by the PWM HW. It should be defined by the HW supplier and used by the PWMDriver to reference specific HW configurations which set the PWM HW module in the referenced power state.	
Screenshot	N/A	
Properties	Property	Value
	Type	Variable: INTEGER
	SymbolicNameValue	True
	Origin	AUTOSAR_ECUC

3.2.6.17.2 PwmPowerStateReadyCbRef

Variable	PwmPowerStateReadyCbRef	
Description	Each instance of this parameter contains a reference to a power mode callback defined in a CDD or IoHwAbs component.	
Screenshot	N/A	
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	SymbolicNameValue	false

	Origin	AUTOSAR_ECUC
--	--------	--------------

3.2.7 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.7.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	4

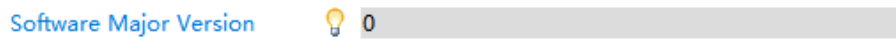
3.2.7.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	6

3.2.7.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.7.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

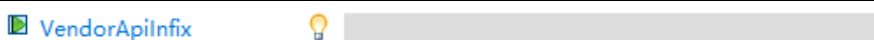
3.2.7.5 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	6

3.2.7.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.7.7 VendorApiInfix

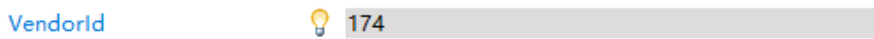
Variable	VendorApiInfix	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable : STRING_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Optional	true

	Readonly	true
--	----------	------

3.2.7.8 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	121

3.2.7.9 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	174

Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

- 1) To configure notification, the user should enable PwmNotificationSupported first and PwmNotification must not be NULL or NULL_PTR.

Pwm

Name  Pwm

General | PwmChannel | PwmFtuModule | PwmEcucPartitionRef | PwmFtuGtbGroup | PwmPowerStateConfig | Published Information

Config Variant  VariantPreCompile 

PwmConfigurationOfOptApiServices

Name  PwmConfigurationOfOptApiServices

PwmDelInitApi  ☒ 	PwmGetOutputState  ☐ 
PwmSetDutyCycle  ☒ 	PwmSetOutputToldle  ☒ 
PwmSetPeriodAndDuty  ☐ 	PwmVersionInfoApi  ☒ 
PwmGetChannelStateApi  ☐ 	PwmEnableTriggerOut  ☐ 

PwmGeneral

Name  PwmGeneral

Pwm Multi-Core Enable  ☒ 	PwmDevErrorDetect  ☐ 
PwmDutyCycleUpdatedEndperiod  ☒ 	PwmNotificationSupported  ☐ 
PwmPeriodUpdatedEndperiod  ☒	☒ PwmLowPowerStatesSupport 
☒ PwmPowerStateAsynchTransitionMode 	Pwm Enable User Mode Support 
☒ PwmKernelEcucPartitionRef 	
PwmIndex (0 -> 4294967295)  0	
Fault Support Enable  ☐ 	EnableMaskingOperations  ☐ 
Pwm Global Time Base Support  ☐ 	

PwmChannel

Name RGB1_BLUE

General PwmChannelEcucPartitionRef

PwmChannelId (0 -> 4294967295) 0

PWM Hardware IP FTU

PwmFtuChannel /Pwm/Pwm/PwmChannelConfigSet/PwmFtuModule_0/RGB1_B

PwmMcuClockReferencePoint

PwmChannelClass

PwmPolarity PWM_HIGH

PwmIdleState PWM_LOW

Default Period(HZ) 10000

Default Period Value(in seconds) (0 -> Infinity) 1.0E-4

PwmDutyCycleDefault (0x0 -> 0x8000) 0x0

PwmNotification Pwm_Notification

- 2) If an even channel exists, such as channel0, and this channel is configured to the combined mode, the channel+1 must not be used, such as channel1.
- 3) If the channel has not been configured to the combined mode, the deadtime function will not be useful. If enabled, the configuration tool will report an error.

PwmFtuChannels

Name RGB1_B

General

Ftu Hardware Channel FTU_CH_1

Edge configuration setting for current channel INDEPENDENT

Enable Deadtime on combined channels* ☒ Enable Channel Phase Shift ☐

4.2 PWM Usage Common Steps

Basically, the PWM module can be configured by following the below 4 steps:

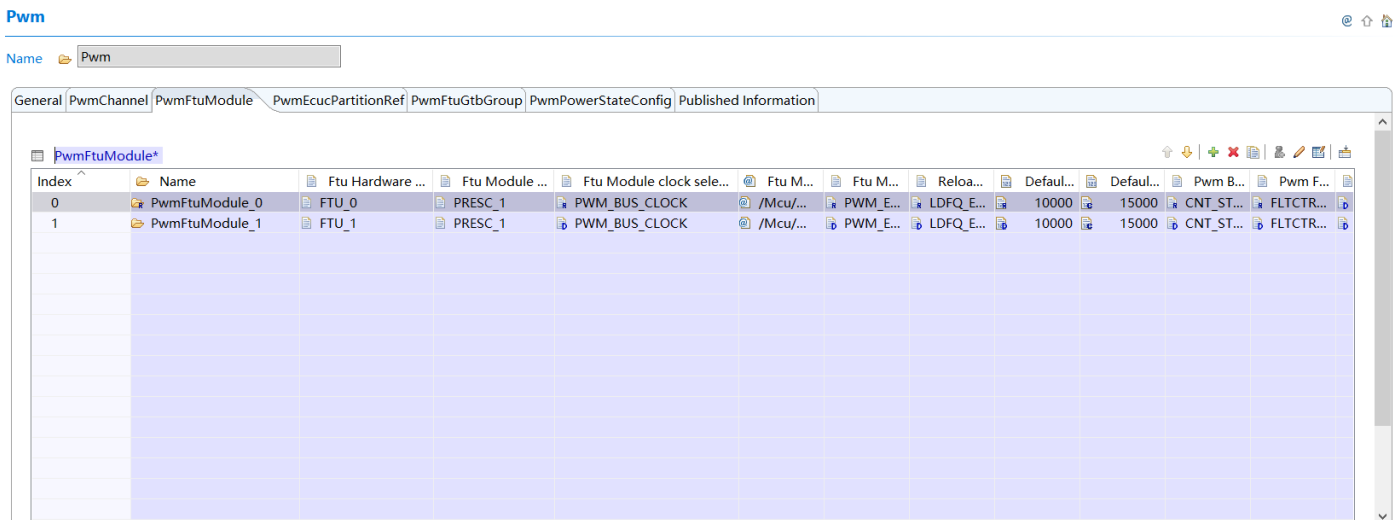
- 1) Configure FTU configurations and FTU channels in the PwmFtuModule tab.
- 2) Configure PWM channel configurations and link the PWM channel to the hardware channel in the PwmFtuModule tab.
- 3) Choose which API will be used.
- 4) Generate configuration files.

4.3 One or Several Independent PWM Channel Demo

- 1) Configure FTU configurations and FTU channels.

In this demo, FTU instance uses FTU0 and FTU1, and adds channel 0 and channel 1 in FTU0 and FTU1. Here are the configurations:

- a) The bus clock source is selected. Therefore, for the clock frequency referenced by the MCU module, need to choose the bus clock reference point.
- b) Select "PRESC_1" for FTU Module Prescaler, so the clock frequency is $\text{PWM_BUS_CLOCK}/1$.
- c) PWM uses edge align mode.
- d) Select "LDFQ_EACH1" for Reload Frequency, so the period and duty value updated at the counter overflows, or the channel value matches the compare value after the `Pwm_SetPeriodAndDuty` or `Pwm_SetDutyCycle` called.
- e) Set the Default Period Value to 10kHz, and the Default Period Value will be calculated automatically.
- f) Fault unused.
- g) External triggers unused.
- h) Use channel 0 and channel 1.
- i) Select INDEPENDENT mode for every channel and disable deadtime.



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- f) Select "PWM_LOW" for PwmIdleState, then the PWM channel output will force to low after calling the API Pwm_SetOutputTolde.
- g) Default Period HZ and Default Period Value will refer to the FTU module configuration.
- h) Set PwmDutycycleDefault to 0x4000, the mean is duty cycle set to 50%.
- i) Do not set notification.
- j) Must set 4 PWM channels and refer to the channels configured before.

PwmChannel

Name: PwmChannel_0

General PwmChannelEcucPartitionRef

PwmChannelId (0 -> 4294967295): 0

PWM Hardware IP: FTU

PwmFtuChannel: /Pwm/Pwm/PwmChannelConfigSet/PwmFtuModule_0/PwmFtuChannels_0

PwmMcuClockReferencePoint: /Mcu/Mcu/McuModuleConfiguration/MCU_Demo_FOSC24M/McuClockReferencePoint_Bus

PwmChannelClass: PWM_FIXED_PERIOD

PwmPolarity: PWM_HIGH

PwmIdleState: PWM_LOW

Default Period(HZ): 10000

Default Period Value(in seconds) (0 -> Infinity): 1.0E-4

PwmDutycycleDefault (0x0 -> 0x8000): 0x4000

Pwm Phase Shift Counter(in seconds) (dynamic range): 0.0

PwmNotification: NULL

Pwm

Name: Pwm

General PwmChannel PwmFtuModule PwmEcucPartitionRef PwmPowerStateConfig Published Information

PwmChannelConfigSet

Name: PwmChannelConfigSet

Ind...	Name	PwmC...	PWM ...	PwmFtuChannel	PwmMcuClockReferencePoint	PwmPolarity	PwmL...	Defaul...	Defaul...	PwmD...
0	PwmChannel_0	0	FTU	/Pwm/Pwm/PwmChannelConfigSet/PwmFtuModule_0/PwmFtuChannels_0	/Mcu/Mcu/McuModuleConfiguration/MCU_D...	PWM_HIGH	PWM_L...	10000	1.0E-4	0x4000
1	PwmChannel_1	1	FTU	/Pwm/Pwm/PwmChannelConfigSet/PwmFtuModule_0/PwmFtuChannels_1	/Mcu/Mcu/McuModuleConfiguration/MCU_D...	PWM_HIGH	PWM_L...	10000	1.0E-4	0x4000
2	PwmChannel_2	2	FTU	/Pwm/Pwm/PwmChannelConfigSet/PwmFtuModule_1/PwmFtuChannels_0	/Mcu/Mcu/McuModuleConfiguration/MCU_D...	PWM_HIGH	PWM_L...	10000	1.0E-4	0x4000
3	PwmChannel_3	3	FTU	/Pwm/Pwm/PwmChannelConfigSet/PwmFtuModule_1/PwmFtuChannels_1	/Mcu/Mcu/McuModuleConfiguration/MCU_D...	PWM_HIGH	PWM_L...	10000	1.0E-4	0x4000

- 3) Configure general configurations.
 - a) Select which API will be used.
 - b) Enable functions of the PWM module, in this demo, enable PwmDutycycleUpdatedEndperiod and the Reload frequency in the PwmFtuModule tab will be useful.

Pwm

Name

General | PwmChannel | PwmFtuModule | PwmEcucPartitionRef | PwmFtuGtbGroup | PwmPowerStateConfig | Published Information

Config Variant

PwmConfigurationOfOptApiServices

Name

PwmDeInitApi	☒		PwmGetOutputState	☐	
PwmSetDutyCycle	☒		PwmSetOutputTolde	☒	
PwmSetPeriodAndDuty	☐		PwmVersionInfoApi	☒	
PwmGetChannelStateApi	☐		PwmEnableTriggerOut	☐	

PwmGeneral

Name

Pwm Multi-Core Enable	☒		PwmDevErrorDetect	☐	
PwmDutyCycleUpdatedEndperiod	☒		PwmNotificationSupported	☐	
PwmPeriodUpdatedEndperiod	☒		PwmLowPowerStatesSupport	☐	
PwmPowerStateAsynchTransitionMode	☐		Pwm Enable User Mode Support	☐	
PwmKernelEcucPartitionRef	☐				
PwmIndex (0 -> 4294967295)	0				
Fault Support Enable	☐		EnableMaskingOperations	☐	
Pwm Global Time Base Support	☐				

- 4) Right-click the module and select Generate Configuration to generate code after the configuration is finished.

4.4 Combined PWM Channel Demo

- 1) Configure FTU configurations and FTU channels.

In this demo, FTU instance use FTU0 and FTU1, and add channel 0 in FTU0 and FTU1. Here are the configurations:

- a) The bus clock source is selected, so the clock frequency reference by the MCU module needs to choose the bus clock reference point.
- b) Select "PRESC_1" for FTU Module Prescaler, so the clock frequency is PWM_BUS_CLOCK/1.
- c) PWM uses edge align mode.
- d) Select "LDFQ_EACH1" for Reload Frequency, so the period and duty value updated at the counter overflows, or the channel value matches the compare value after the Pwm_SetPeriodAndDuty or Pwm_SetDutyCycle called.
- e) Set the Default Period Value to 10khz and the Default Period Value will be calculated automatically.
- f) Fault unused.
- g) External triggers unused.
- h) Use channel 0 (if the combined mode is used, must be the even channel).
- i) Select COMBINED_SYNCED mode for the channel and disable deadtime.
- j) (COMBINED_SYNC mode: if channel0 configured, the channel1 will generate PWM signal same as channel0; COMBINED_COMPLEMENTARY mode: if channel0 configured, the channel1 will generate PWM signal complement with channel1)





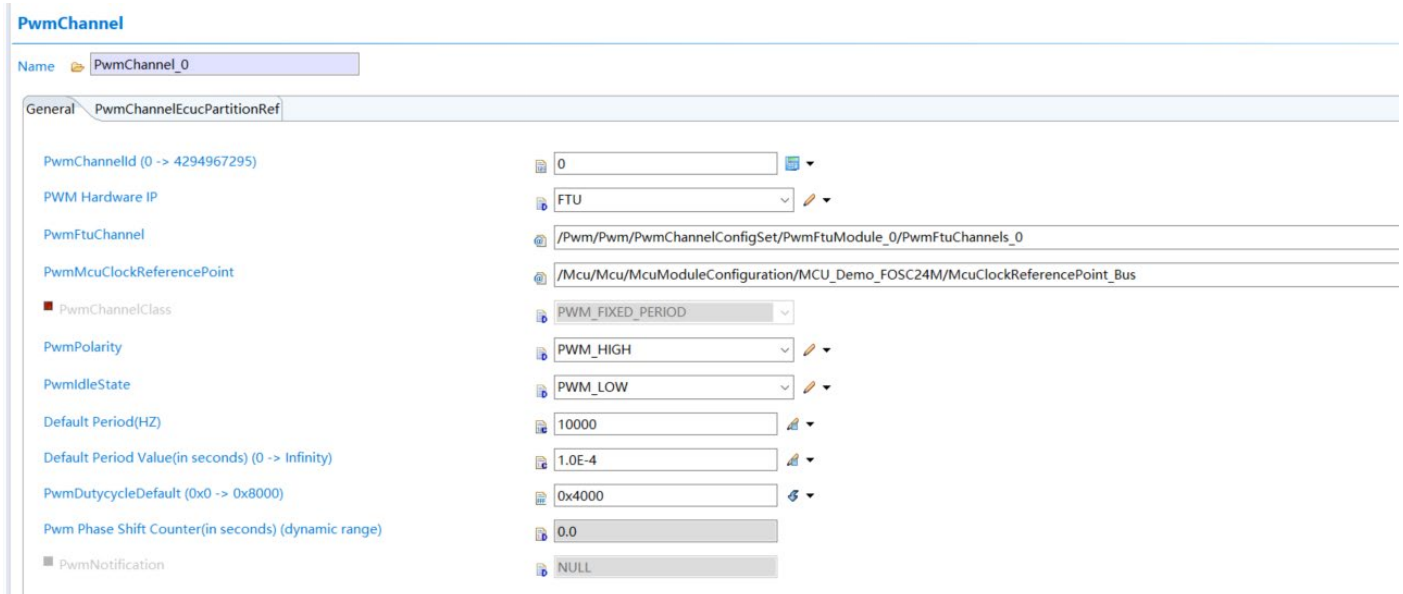
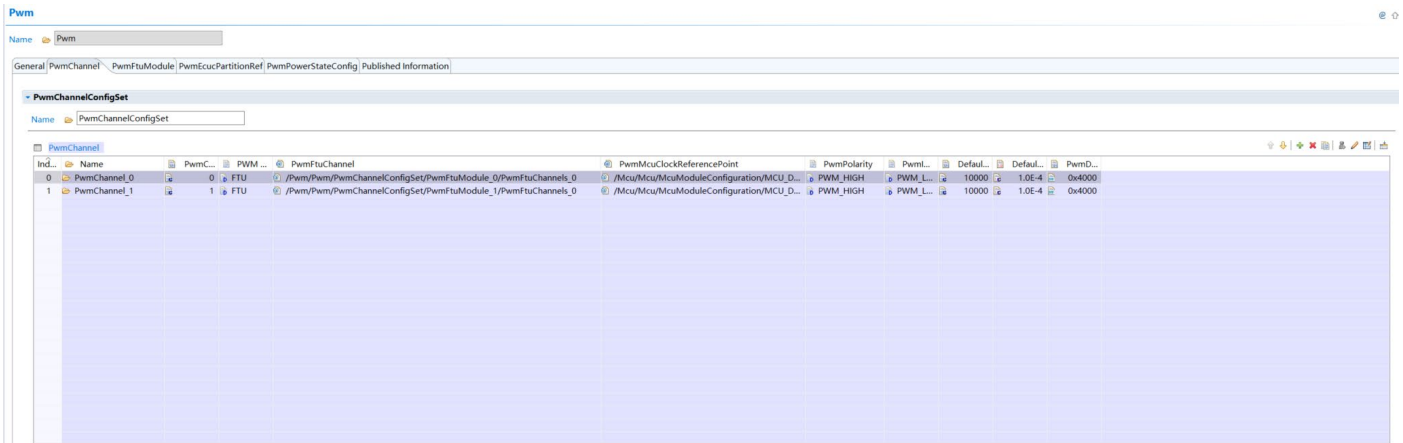






2) Configure the PWM channel.

- a) PWM channel hardware uses FTU (In FC7XXX series chip, PWM only can use FTU module).
- b) Link to the channel configured before.
- c) Select the same clock source as the PwmFtuModule to which the current channel belongs.
- d) Select "PWM_FIXED_PERIOD" for PwmChannelClass. In this mode, the period of this channel will not be changed.
- e) Select "PWM_HIGH" for PwmPolarity.
- f) Select "PWM_LOW" for PwmIdleState, then the PWM channel output will force to low after calling the API Pwm_SetOutputToldle.
- g) Default Period HZ and Default Period Value will refer to the FTU module configuration.
- h) Set PwmDutycycleDefault to 0x4000, the mean is duty cycle set to 50%.
- i) Do not set notification.
- j) Must set 2 PWM channels and refer to the channels configured before.



- 3) Configure general configurations.
 - a) Select which API will be used.
 - b) Enable functions of the PWM module, in this demo, enable PwmDutycycleUpdatedEndperiod and the Reload frequency in PwmFtuModule tab will be useful.

Pwm

Name

General | PwmChannel | PwmFtuModule | PwmEcucPartitionRef | PwmFtuGtbGroup | PwmPowerStateConfig | Published Information

Config Variant

PwmConfigurationOfOptApiServices

Name

PwmDeInitApi	☒		PwmGetOutputState	☐	
PwmSetDutyCycle	☒		PwmSetOutputTolde	☒	
PwmSetPeriodAndDuty	☐		PwmVersionInfoApi	☒	
PwmGetChannelStateApi	☐		PwmEnableTriggerOut	☐	

PwmGeneral

Name

Pwm Multi-Core Enable	☒		PwmDevErrorDetect	☐	
PwmDutycycleUpdatedEndperiod	☒		PwmNotificationSupported	☐	
PwmPeriodUpdatedEndperiod	☒		PwmLowPowerStatesSupport	☐	
☐ PwmPowerStateAsynchTransitionMode			Pwm Enable User Mode Support	☐	
☐ PwmKernelEcucPartitionRef					
PwmIndex (0 -> 4294967295)	0				
Fault Support Enable	☐		EnableMaskingOperations	☐	
Pwm Global Time Base Support	☐				

- 4) Right-click the module and select Generate Configuration to generate code after configuration is finished.